



Chapter 2 — Basic MOS Device Physics

Design of Analog CMOS
Integrated circuits - Behzad
Razavi

Presented by - Chinmay Joshi



CHAPTER OVERVIEW

Concepts covered -

1. MOSFET Structure
2. MOS Regions
3. MOS I/V characteristics
4. Second Order effects
5. MOS Capacitances
6. MOS small signal model

Problem 2.1

2.1. For $W/L = 50/0.5$, plot the drain current of an NFET and a PFET as a function of $|V_{GS}|$ as $|V_{GS}|$ varies from 0 to 3 V. Assume that $|V_{DS}| = 3$ V.

Concepts covered -

1. MOS Regions
 - a. Cut off ($V_{GS} < V_{th}$) - also covered subthreshold conduction
 - b. Triode / linear ($V_{DS} < V_{GS} - V_{th}$)
 - c. Saturation ($V_{DS} > V_{GS} - V_{th}$)
2. Dependence of Drain current (I_D) on Gate-Source Voltage (V_{GS})
 - a. Cut off - $I_D \sim 0$ (I_D depends on $\exp(V_{GS})$)

b. Triode - NMOS

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

PMOS

$$I_D = -\mu_p C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

c. Saturation -NMOS

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L'} (V_{GS} - V_{TH})^2$$

PMOS

$$I_D = -\frac{1}{2} \mu_p C_{ox} \frac{W}{L'} (V_{GS} - V_{TH})^2$$

Handwritten Solution NMOS

Points which were noted -

- The subthreshold current was not taken into consideration while drawing the graph as it is almost negligible (shown in simulation).
- NMOS is always in saturation region in this question.

2.1 Given

$$\frac{W}{L} = \frac{50}{0.5}, V_{T0} = 0.7V, |V_{DS}| = 3V$$

17 Plot I_D v/s $V_{GS} \in (0, 3V)$

Assumption = $\lambda = 0, V_{SB} = 0$

$\therefore$ in cut off

$$V_{GS} < V_{TH} = 0.7V$$

$$I_D \approx 0$$

$\therefore$ after $V_{GS} > V_{TH}$

for saturation

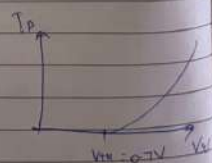
$$V_{GS} > V_{GS} - V_T$$

according to given values

$$3 > 3(V_{GSmax}) - 0.7 \rightarrow \text{always true}$$

$\therefore$ MOSFET is in saturation region

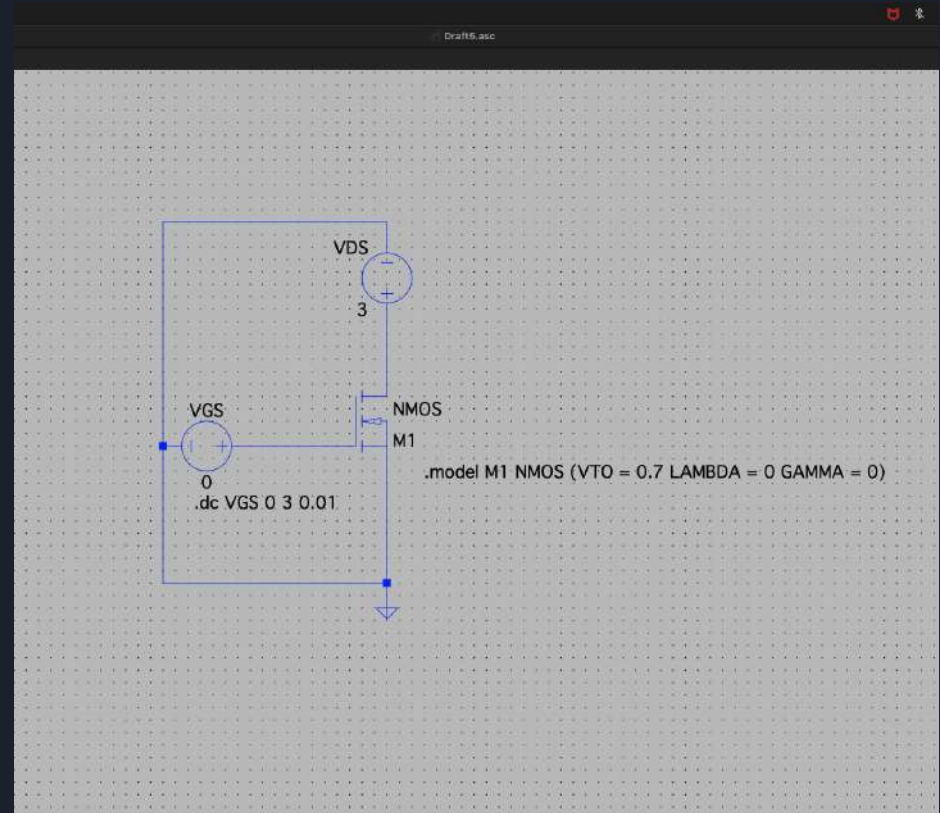
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

$$\therefore I_D \propto (V_{GS} - V_{TH})^2$$


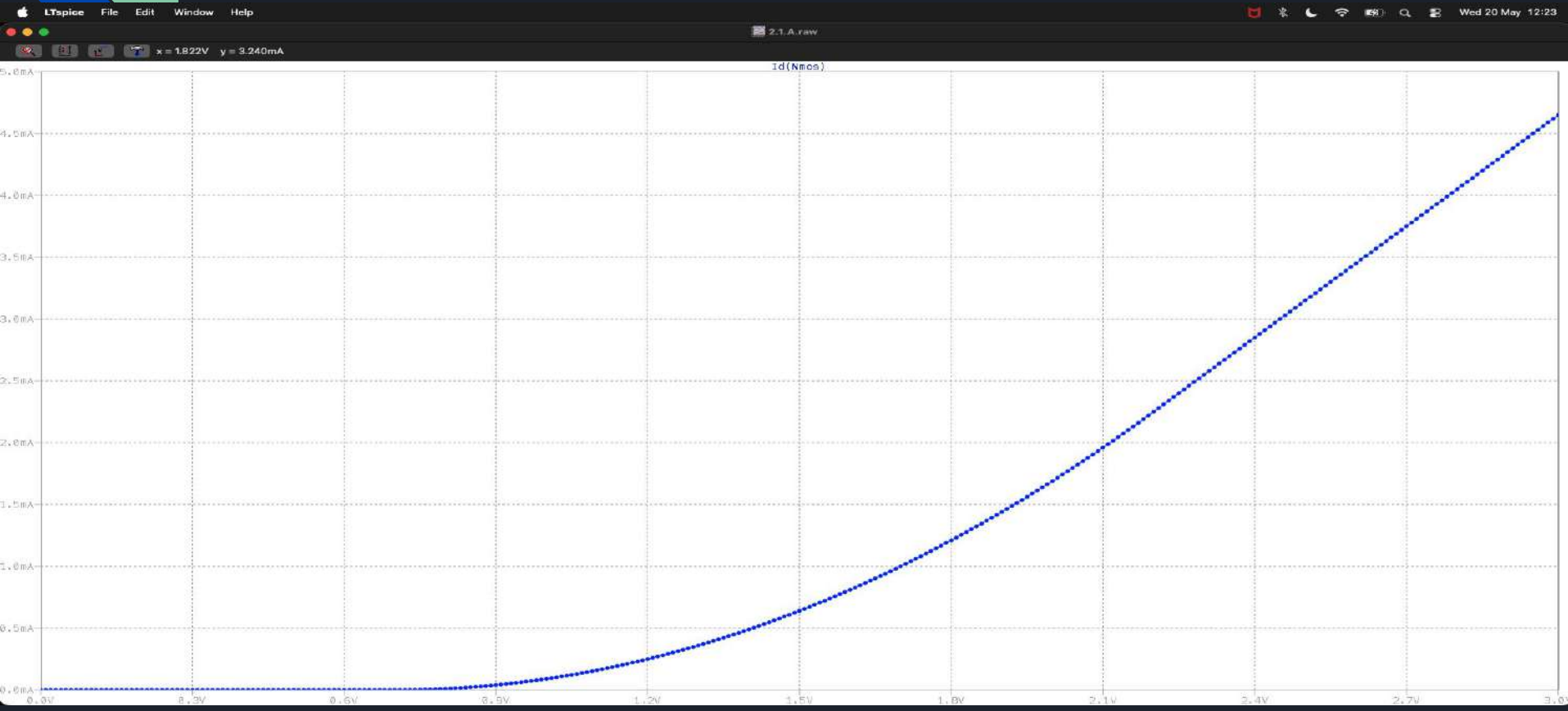
LTSPICE Schematic for NMOS

Points which were noted -

- Spice Directive `.dc VGS 0 3 0.01` was used to sweep V_{GS} from 0 V to 3 V with 0.01 V increments.
- `.model M1 NMOS (VTO = 0.7 LAMBDA = 0 GAMMA = 0)` was used to define the MOSFET threshold voltage and to eliminate body effect and channel-length modulation for idealized analysis.



Simulation Results for NMOS



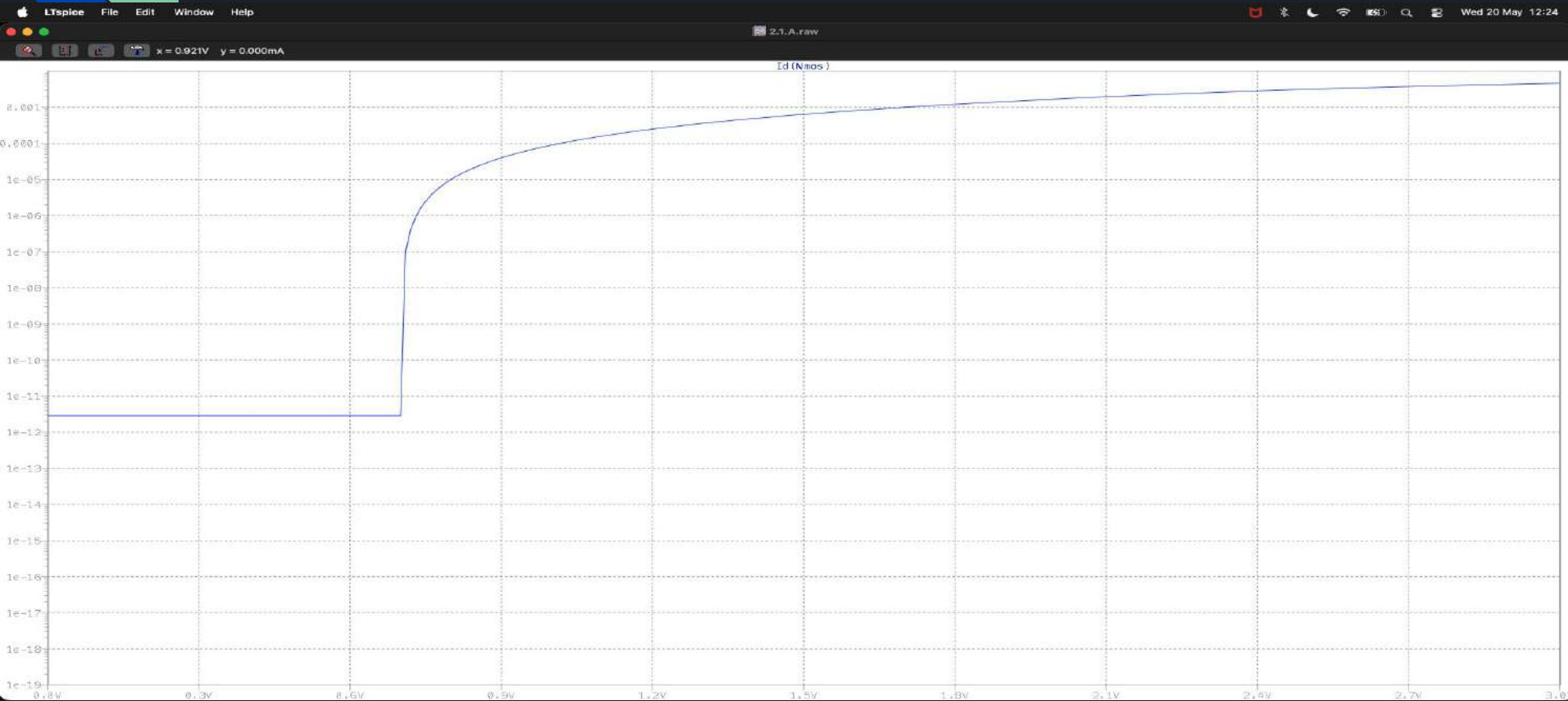


Simulation Results for NMOS

Result analysis -

- The drain current I_D is nearly zero for $V_{GS} < V_{TH}$. For $V_{GS} > V_{TH}$, the MOSFET operates in saturation due to $V_{DS} = 3V$, and the drain current approximately follows the square-law characteristic.
- Further, the Y-axis was set to logarithmic scale to analyze the subthreshold region of operation. In this region, the drain current varies approximately exponentially with gate-source voltage:

Simulation Results with logarithmic Y-axis



OBSERVATION

Parameter	Theoretical Expectation	LTspice Observation
Cutoff Region	$I_D \approx 0$ for $V_{GS} < V_{TH}$	Drain current remained nearly zero below threshold voltage
Threshold Voltage	MOSFET turns ON near $V_{TH} = 0.7V$	Current started increasing near $0.7V$
Operating Region	Saturation for most of sweep range	MOSFET remained in saturation due to $V_{DS} = 3V$
Current Relation	Square-law dependence in strong inversion	I_D increased rapidly for $V_{GS} > V_{TH}$
Subthreshold Region	Exponential dependence of I_D on V_{GS}	Exponential variation observed on logarithmic scale

Handwritten Solution PMOS

Points which were noted -

- The subthreshold current was not taken into consideration while drawing the graph as it is almost negligible (shown in simulation).
- PMOS is always in saturation region in this question.

solution for PMOS

Given $\frac{W}{L} = \frac{50}{0.5}$, $V_{SD} = 3V$, $V_{th} = -0.7V$

in cutoff

$$|V_{GS}| < |V_{th}| = 0.7$$
$$I_D = 0$$

after $|V_{GS}| > 0.7$

for saturation

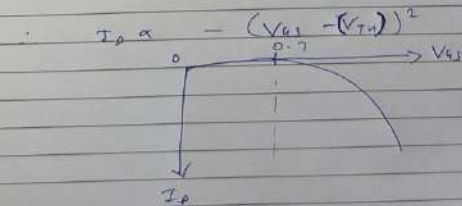
$$V_{SD} \geq V_{SG} - |V_{th}|$$
$$3 \geq V_{SG} - 0.7, \quad V_{SG} \in (0, 3)$$

↓
always true

∴ MOSFET in saturation

$$I_D = -\frac{1}{2} \mu_{eff} \frac{W}{L} (V_{GS} - V_{th})^2$$

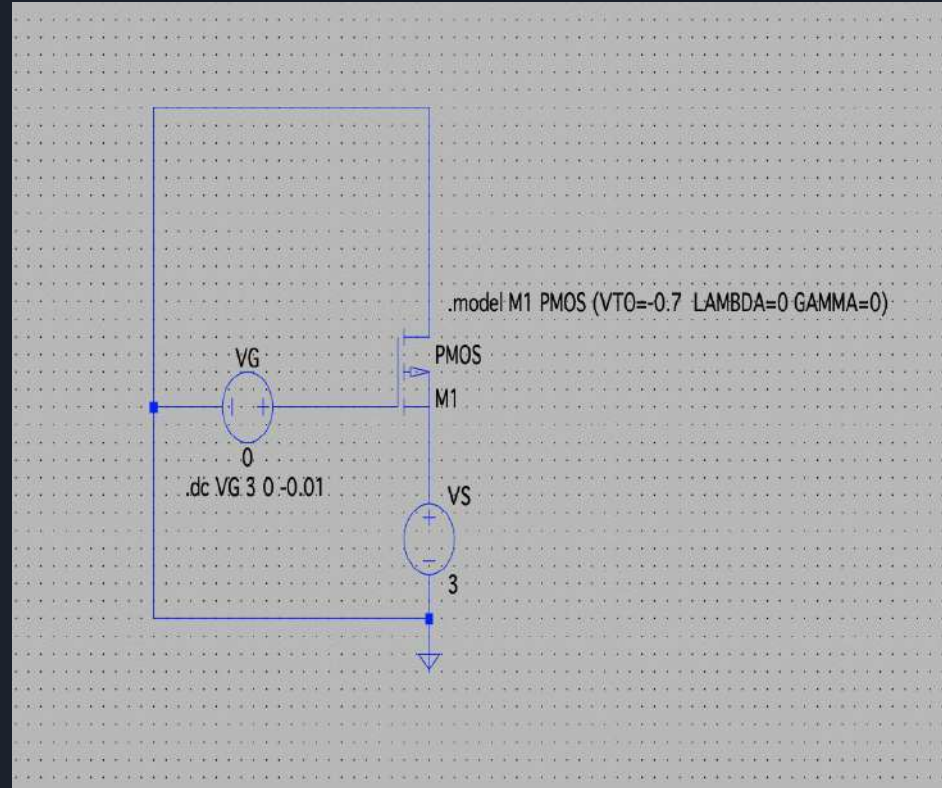
∴ $I_D \propto - (V_{GS} - V_{th})^2$



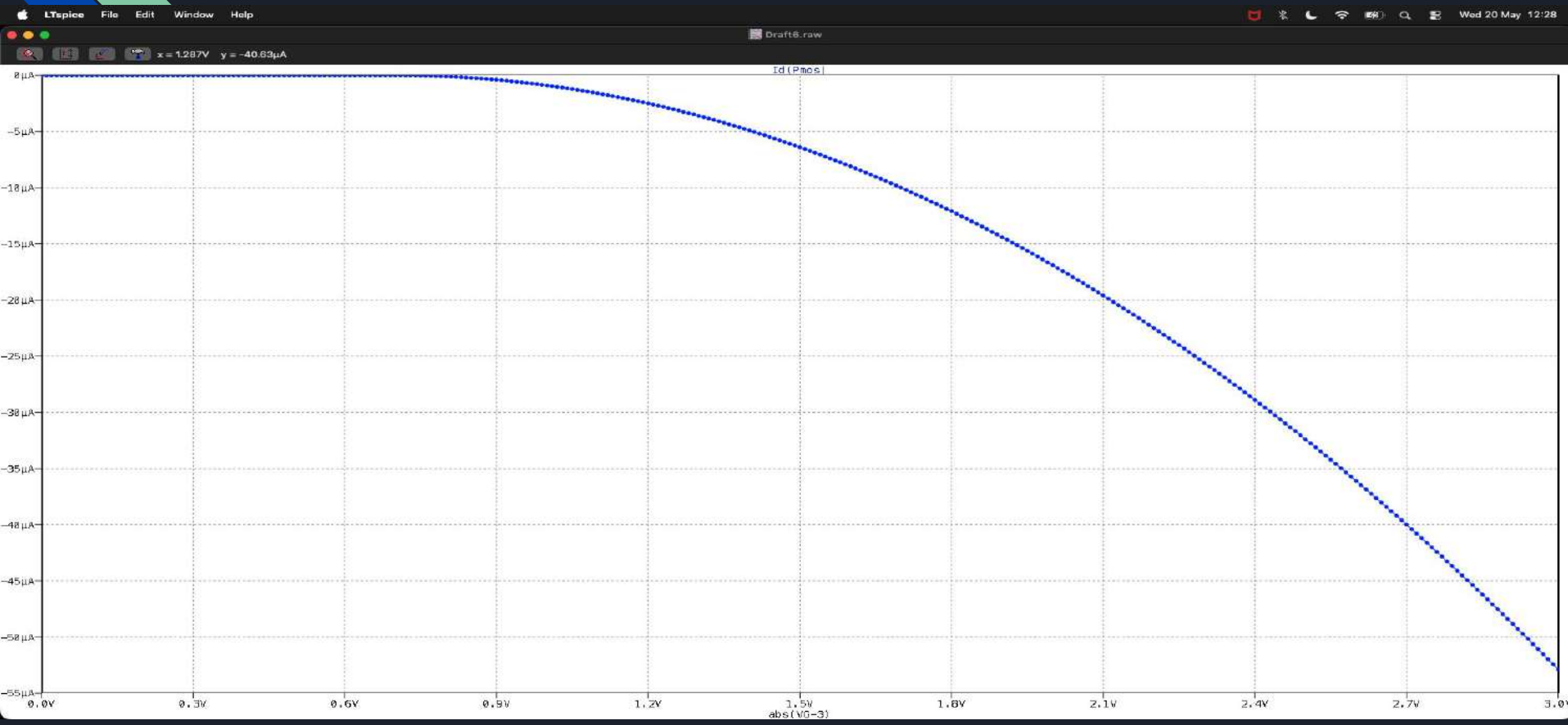
LTSPICE Schematic for PMOS

Points which were noted -

- Spice Directive `.dc VG 3 0 -0.01` was used to sweep V_G from 3 V to 0 V with 0.01 V decrements.
- `.model M1 PMOS (VTO = -0.7 LAMBDA = 0 GAMMA = 0)` was used to define the MOSFET threshold voltage and to eliminate body effect and channel-length modulation for idealized analysis.



Simulation Results for PMOS



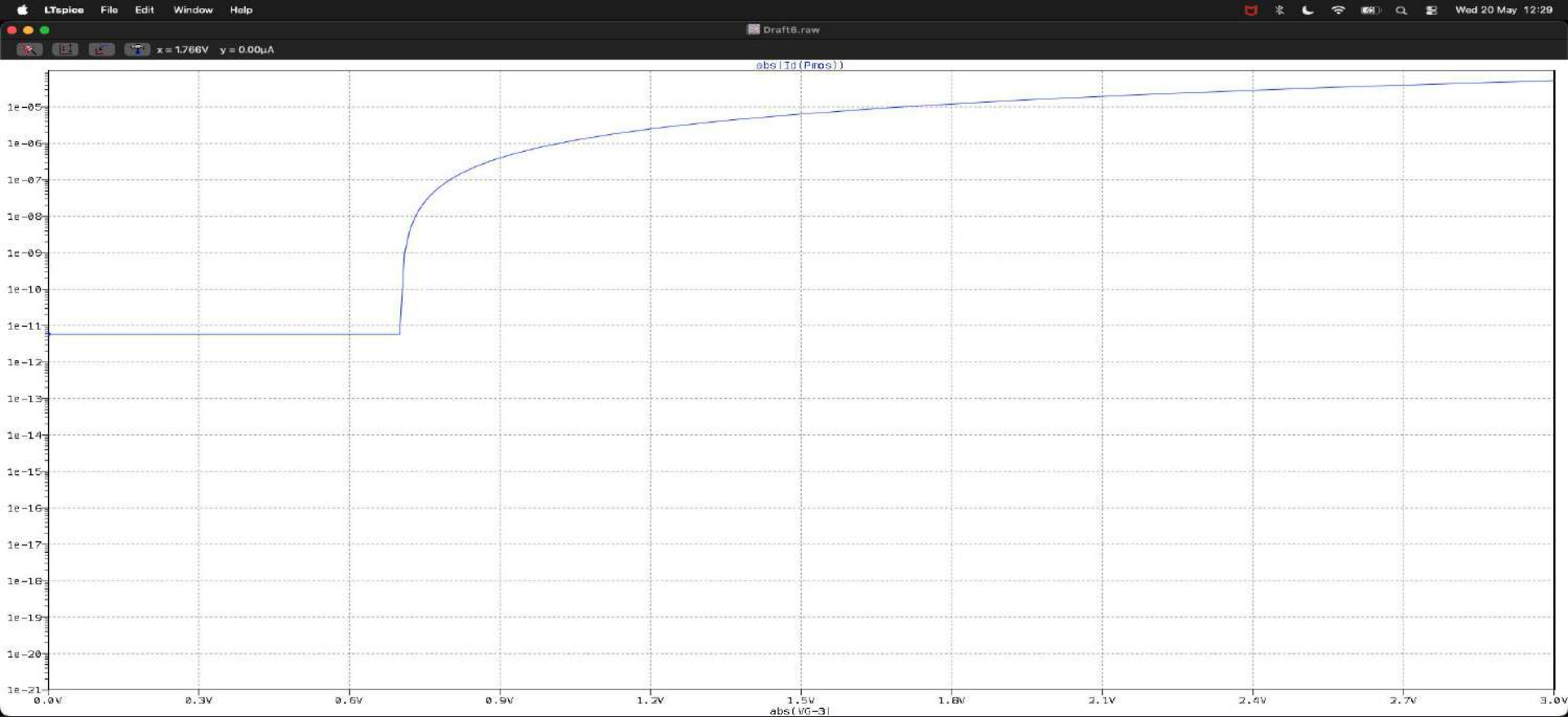


Simulation Results for NMOS

Result analysis -

- The drain current I_D is nearly zero for $V_{GS} < |V_{TH}|$. For $V_{GS} > |V_{TH}|$, the MOSFET operates in saturation due to $V_{DS} = -3V$, and the drain current approximately follows the square-law characteristic.
- Further, the Y-axis was set to logarithmic scale for the absolute value of I_D (as negative current values cannot be plotted directly on a logarithmic scale) to analyze the subthreshold region of operation. In this region, the drain current varies approximately exponentially with gate-source voltage.

Simulation Results with logarithmic Y-axis



OBSERVATION

Parameter	Theoretical Expectation	LTspice Observation
Cutoff Region	$I_D \approx 0$ for $ V_{GS} < V_{TH} $	Drain current remained nearly zero below threshold voltage
Threshold Voltage	MOSFET turns ON near $ V_{TH} = 0.7V$	Current started increasing near $V_{SG} = 0.7V$
Operating Region	Saturation for most of sweep range	MOSFET remained in saturation due to $V_{SD} = 3V$
Current Relation	Square-law dependence in strong inversion	$ I_D $ increased rapidly for $ V_{GS} > V_{TH} $
Subthreshold Region	Exponential dependence of I_D on V_{GS}	Exponential variation observed on logarithmic scale

Problem 2.5-a (Assuming $V_{SB}=0$)

- 2.5. Sketch I_X and the transconductance of the transistor as a function of V_X for each circuit in Fig. 2.47 as V_X varies from 0 to V_{DD} . In part (a), assume that V_X varies from 0 to 1.5 V.

Diagram given -

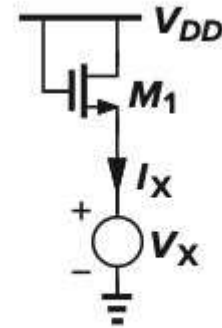
Formula used -

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L'} (V_{GS} - V_{TH})^2$$

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS} \text{ const.}}$$

$$= \frac{2I_D}{V_{GS} - V_{TH}}$$



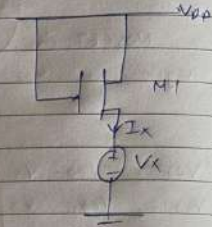
(a)

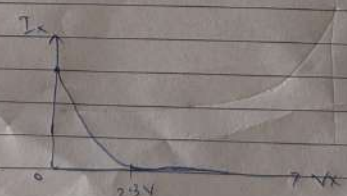
Handwritten Solution

Points which were noted -

- $I_x = I_d$
- Once V_x crosses 2.3 V, V_{gs} becomes less than V_{th} i.e $V_{gs} < V_{th}$, therefore MOSFET enters cut-off region and current becomes zero. ($I_d = I_x = 0$).

2.5.4 a-y Given



$V_{gs} = V_{DD} - V_x = 3 - V_x$
 $V_{ds} = V_{DD} - V_x = 3 - V_x$
 $\therefore V_{gs} = V_{ds}$
 MOSFET in saturation
 if when $V_x < 2.3V$
 MOSFET in saturation
 $I_x = I_d = \frac{1}{2} K_n' \frac{W}{L} (V_{gs} - V_{th})^2$
 $= \frac{1}{2} K_n' \frac{W}{L} (3 - V_x - 0.7)^2$
 $I_x = I_d = \frac{1}{2} K_n' \frac{W}{L} (2.3 - V_x)^2$
 2.5 when $V_x > 2.3V$, suppose 2.4 V
 $V_{gs} = 3 - 2.4 = 0.6 < V_{th}$
 $\therefore$ MOSFET gets cut off


Handwritten Solution

Points which were noted -

- Since MOSFET enters cut-off for $V_x > 2.3V$.
- Transconductance also becomes zero. ($g_m = 0$)

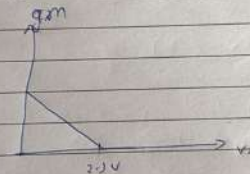
ii) Trans conductance

a.y $V_x < 2.3V$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \frac{\partial I_D}{\partial V_x}$$
$$I_D = \frac{1}{2} K_n' \frac{W}{L} (V_{GS} - V_{th})^2$$
$$g_m = K_n' \frac{W}{L} (V_{GS} - V_{th})$$
$$\therefore g_m \propto (V_{GS} - V_{th})$$
$$g_m \propto (3 - V_x - 0.7)$$
$$g_m \propto (2.3 - V_x)$$

a.y After $V_x = 2.3V$

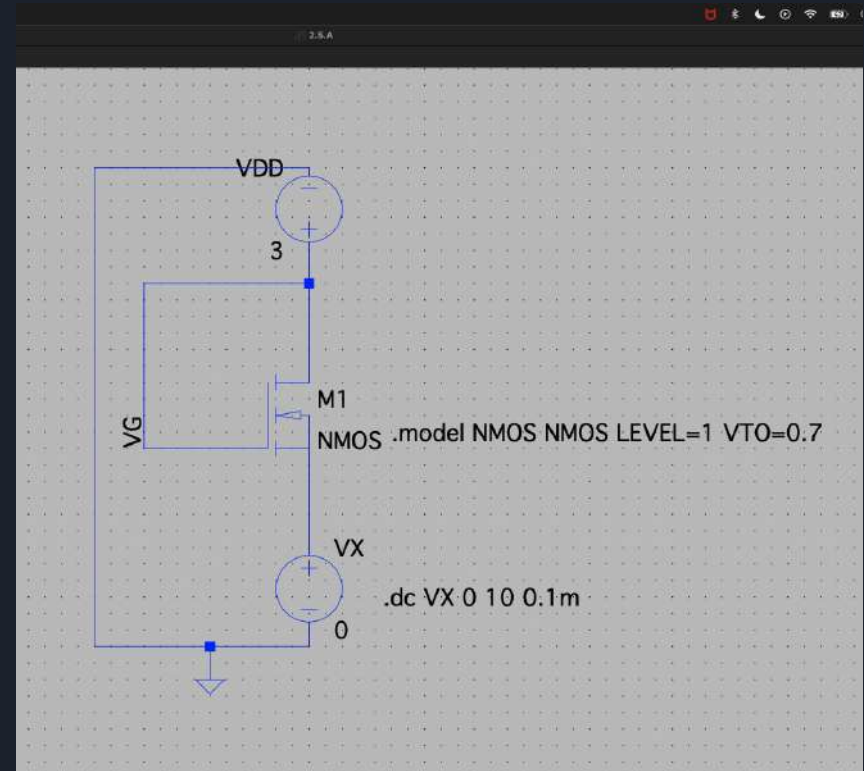
MOSFET cut-off

$$g_m = 0$$


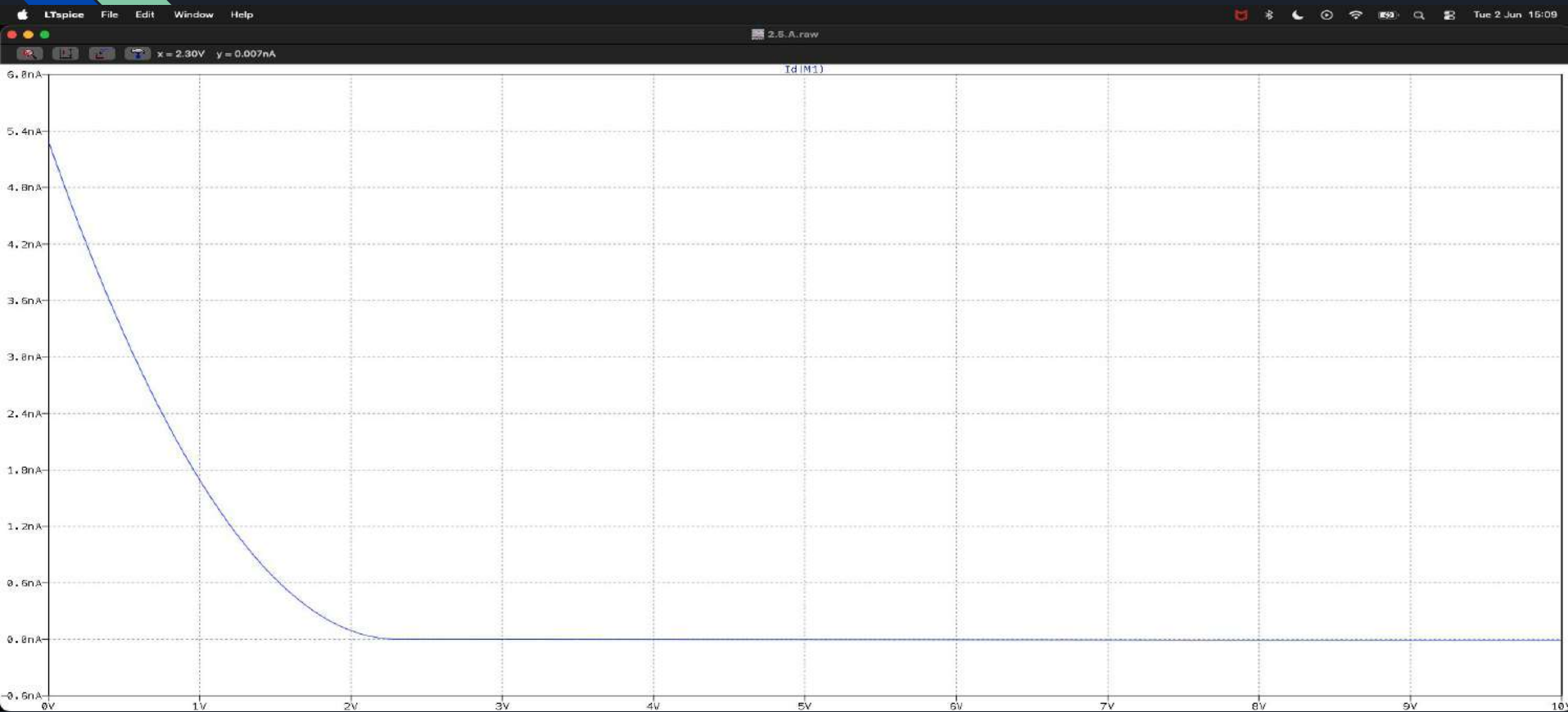
LTSPICE Schematic

Points which were noted -

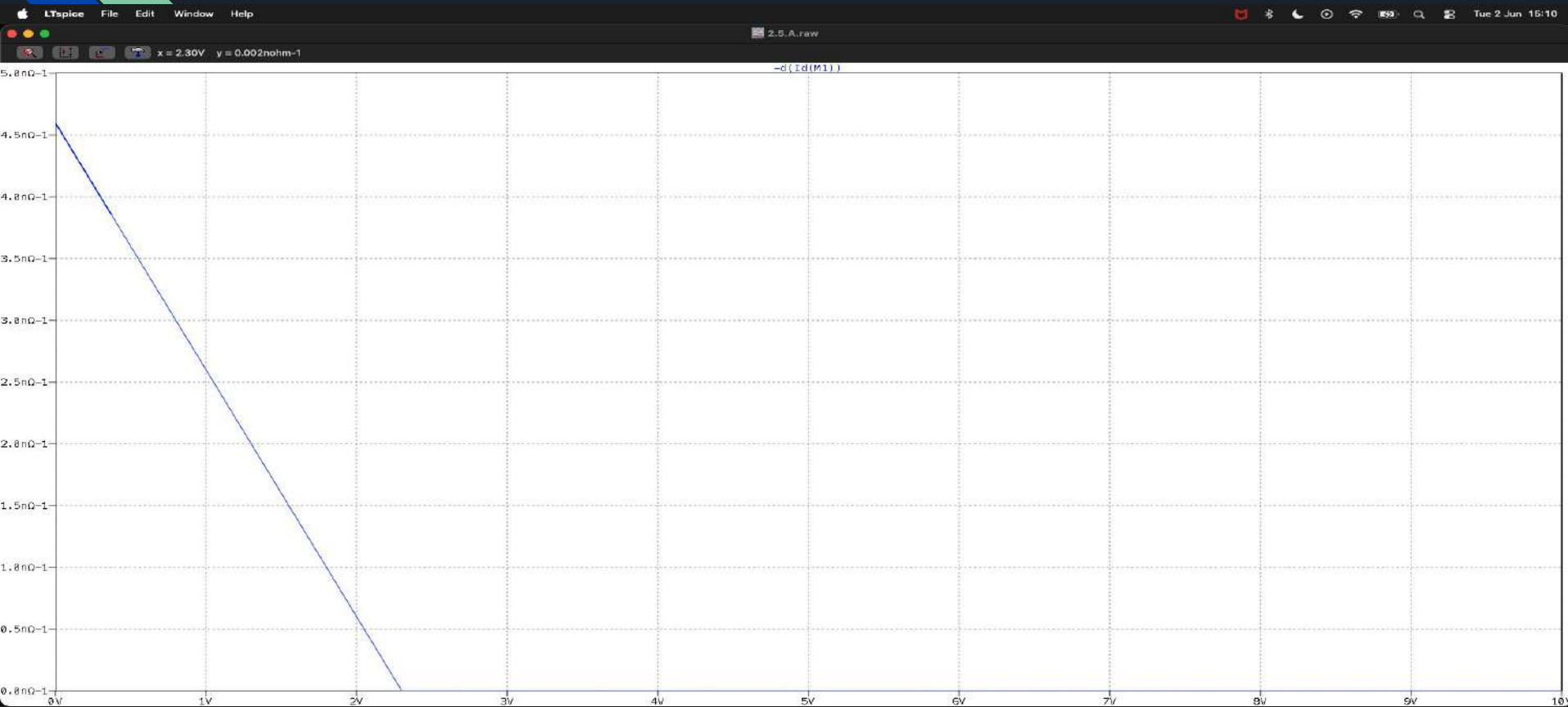
- Spice Directive `.dc VX 0 3 0.1m` was used to sweep VX from 0V to 10V with 1 mV increments.
- `.model NMOS NMOS (LEVEL 1 VTO = 0.7)` was used to model the MOSFET, LEVEL=1 selects the basic Shichman-Hodges MOSFET model, while VTO=0.7 sets the threshold voltage of the n-channel MOSFET to 0.7 V
-



Simulation Results for I_x



Simulation Results for gm





OBSERVATION

- When the gate and drain terminals of an n-channel MOSFET are connected together, the device operates in the saturation region as long as $V_{GS} > V_{TH}$.
- The drain current I_D decreases exponentially with an increase in V_X and becomes approximately zero at $V_X = 2.3V$.
- The transconductance g_m decreases nearly linearly as V_X increases and reaches zero at $V_X = 2.3V$.
- For $V_X > 2.3V$, the MOSFET turns off since V_{GS} falls below the threshold voltage, resulting in both I_D and g_m being zero.

Problem 2.5-a (considering $V_{SB} = V_X$)

2.5. Sketch I_X and the transconductance of the transistor as a function of V_X for each circuit in Fig. 2.47 as V_X varies from 0 to V_{DD} . In part (a), assume that V_X varies from 0 to 1.5 V.

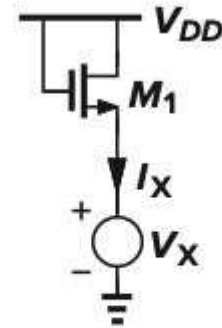
Diagram given -

Formula used -

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \quad I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS} \text{ const.}} = \frac{2I_D}{V_{GS} - V_{TH}}$$

$$V_{TH} = V_{TH0} + \gamma \left(\sqrt{2\Phi_F + V_{SB}} - \sqrt{2\Phi_F} \right)$$



(a)

Handwritten Solution

Points which were noted -

- $I_x = I_d$
- Since $V_{GS} = V_{DS}$, MOSFET is always in saturation region, we only need to check for $V_{GS} > V_{TH}$

2.5 Ya 9 Given

$V_{DS} = V_{GS} = 3 - V_x$

Since $V_{GS} = V_{DS}$, MOSFET always in saturation if $V_{GS} > V_{th}$

$\therefore$ for $V_{GS} > V_{th}$

$3 - V_x > 0.7 + 0.4 \left(\sqrt{0.9 + V_x} - \sqrt{0.9} \right), V_{th} = V_T$

Let $\sqrt{0.9 + V_x} = y$
 $V_x = y^2 - 0.9$

$3 - y^2 + 0.9 > 0.7 + 0.4 \left(y - \sqrt{0.9} \right)$

$0 > y^2 - 3.9 + 0.7 + 0.4 \sqrt{y} - 0.4269$

$0 > y^2 + 0.4 \sqrt{y} - 3.6269$

$\therefore y = 1.696$

$\therefore V_x = y^2 - 0.9 = 1.976 \text{ V}$

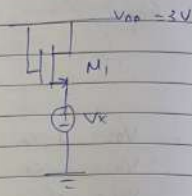
$\therefore I_x = I_d = \frac{1}{2} \mu_n C_{ox} \left(3 - V_x - 0.7 - 0.4 \left(\sqrt{0.9 + V_x} - \sqrt{0.9} \right) \right)^2$

h.u $V_x = 1.976 \text{ V}$

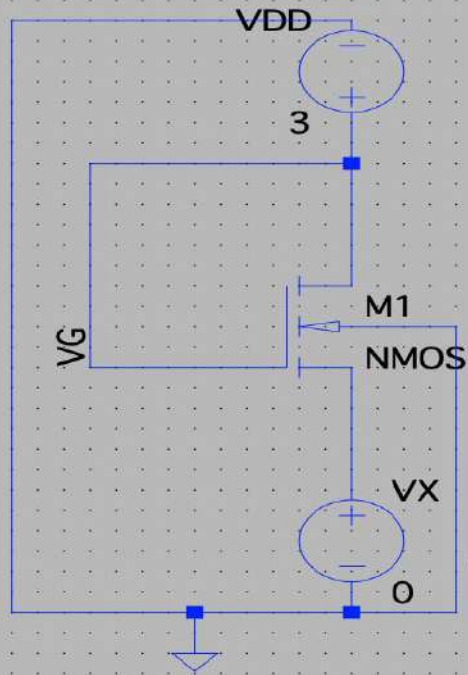
for $V_x > 1.976 \text{ V}, V_{GS} < V_{th}$

MOSFET cutoff

$g_m = \frac{\partial I_d}{\partial V_{GS}} \bigg|_{V_{DS} = \text{const}} = \sqrt{2 \mu_n C_{ox} I_D} = \sqrt{2 \mu_n C_{ox} I_x}$



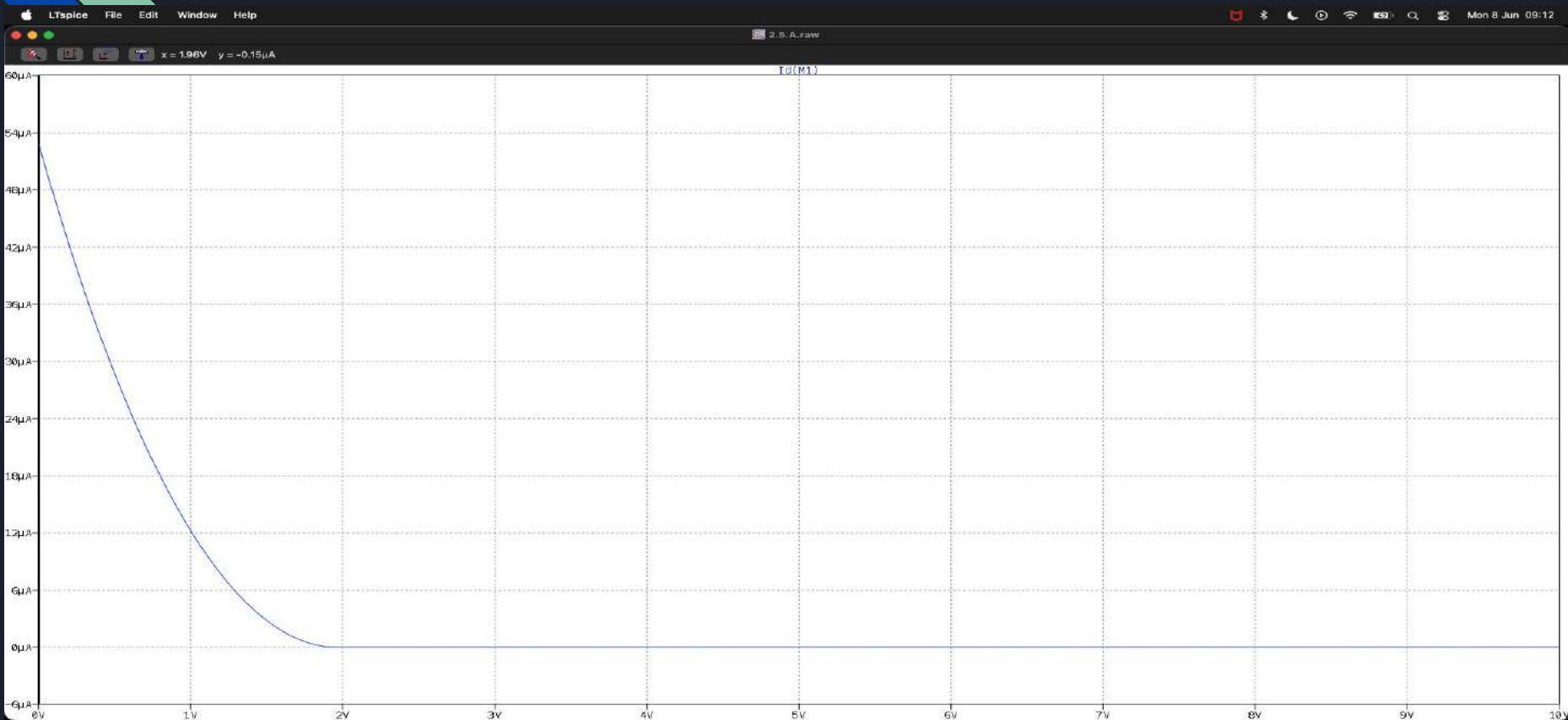
LTSPICE Schematic



.model NMOS NMOS LEVEL=1 VTO=0.7 GAMMA = 0.45 PHI = 0.9

.dc VX 0 10 0.1m

Simulation Results for I_x



Problem 2.5-b

- 2.5. Sketch I_X and the transconductance of the transistor as a function of V_X for each circuit in Fig. 2.47 as V_X varies from 0 to V_{DD} . In part (a), assume that V_X varies from 0 to 1.5 V.

Diagram given -

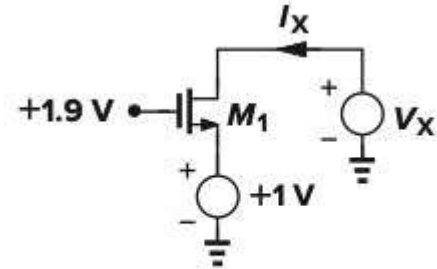
Formula used -

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L'} (V_{GS} - V_{TH})^2$$

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS} \text{ const.}}$$

$$= \frac{2I_D}{V_{GS} - V_{TH}}$$

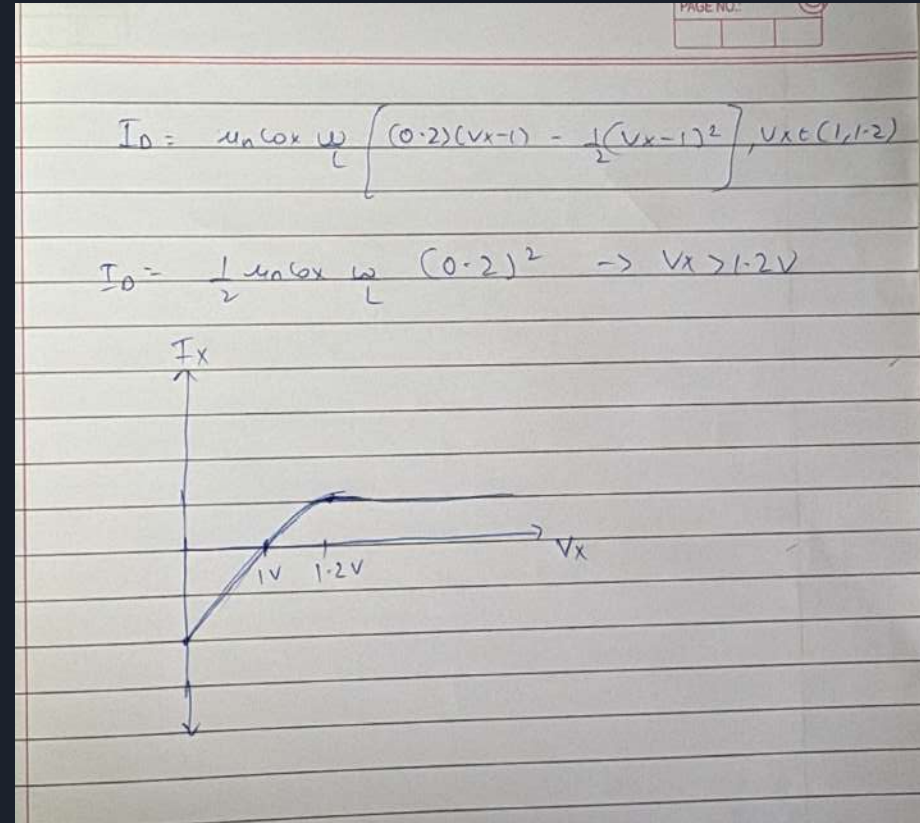


(b)

Handwritten Solution

Points which were noted -

- For $V_X < 1V$, the drain and source are interchanged.
- After $V_X = 1V$ the direction of current reverses and the MOSFET is in linear region till $V_X = 1.2V$
- After $V_X = 1.2V$, MOSFET enters saturation region.

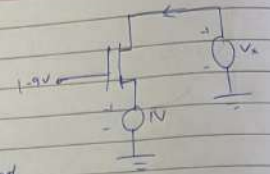


Handwritten Solution

Points which were noted -

- For $V_X < 1V$, the drain and source are interchanged.
- After $V_X = 1V$ the direction of current reverses and the MOSFET is in linear region till $V_X = 1.2V$
- After $V_X = 1.2V$, MOSFET enters saturation region.

2546y



Case 1 $\therefore$ till $V_X = 1V$

Drain & S are interchanged

$$V_{GS} = 1.9 - V_X, \quad V_{DS} = 1 - V_X, \quad V_{GS} - V_{th} = 1.2 - V_X$$

$\therefore$ For MOSFET to turn ON

$$V_{GS} > V_{th}$$

$$1.9 - V_X > 0.7$$

$$1.2 - V_X > 0$$

$$V_X < 1.2$$

$\therefore$ MOSFET is ON till $V_X = 1V$

Let us consider

$$V_X = 0, \quad V_{GS} - V_{th} = 1.2V, \quad V_{DS} = 1V \rightarrow \text{linear}$$

$$V_X = 1, \quad V_{GS} - V_{th} = 0.2, \quad V_{DS} = 0 \rightarrow \text{linear}$$

$$\therefore I_X = I_D = -\frac{\mu_n C_{ox} W}{L} \left[\frac{(1.2 - V_X)(1 - V_X)}{2} - \frac{1}{2}(1 - V_X)^2 \right]$$

Case 2 After $V_X = 1V$

direction of current reverses

$$V_{GS} = 0.9V, \quad V_{DS} = V_X - 1, \quad V_{GS} - V_{th} = 0.2V$$

$\therefore$ for triode

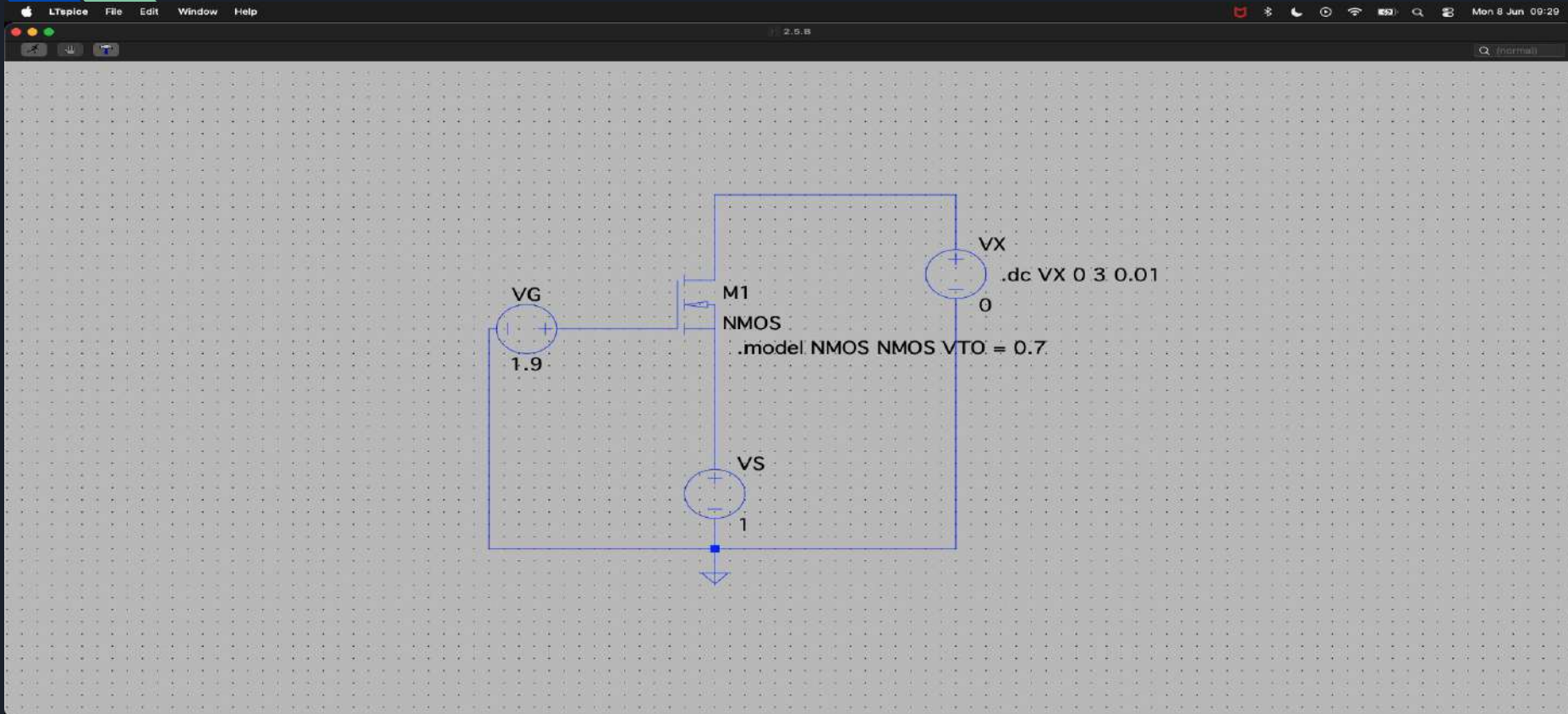
$$V_{DS} < V_{GS} - V_{th}$$

$$V_X - 1 < 0.2$$

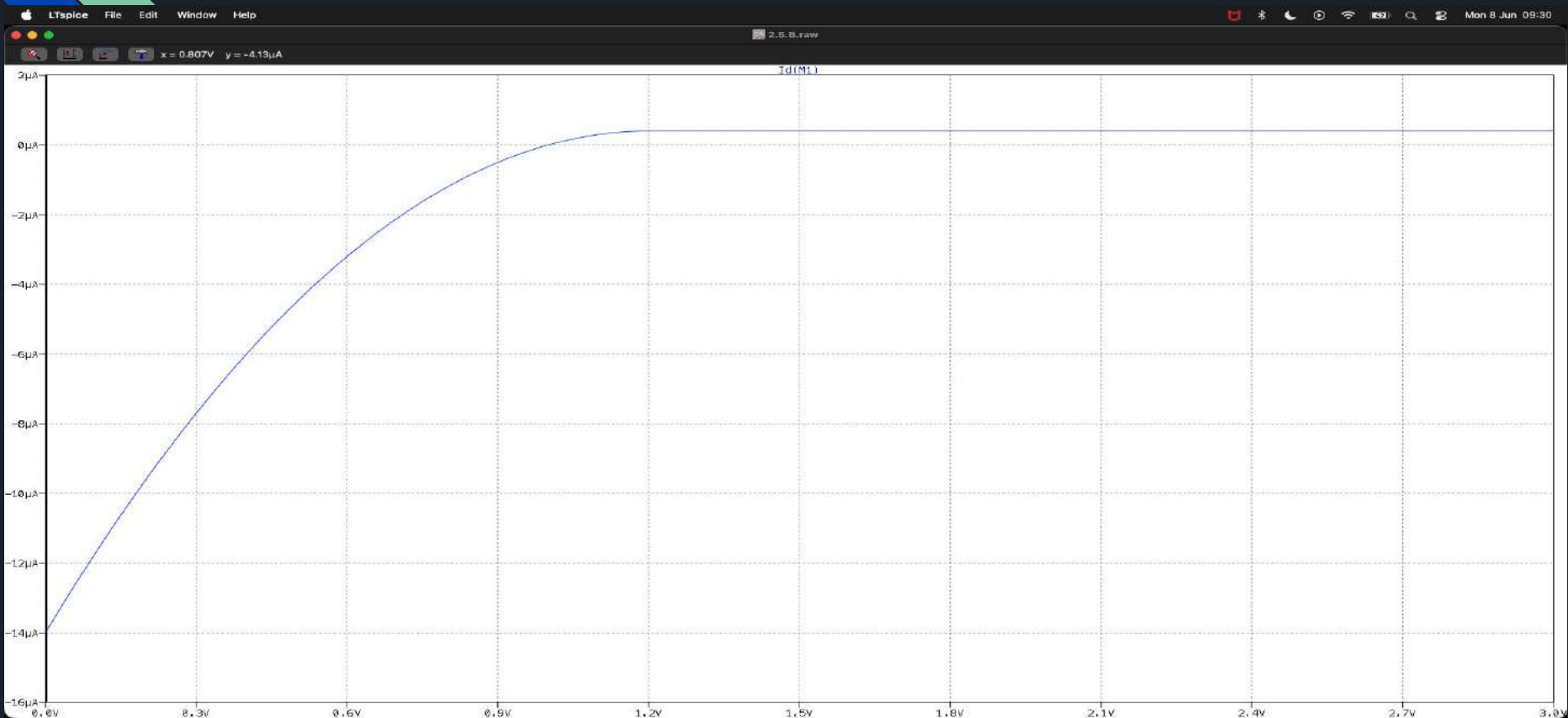
$$V_X < 1.2V \rightarrow \text{triode}$$

$$V_X > 1.2V \rightarrow \text{saturation}$$

LTSPICE Schematic



Simulation Results for I_x





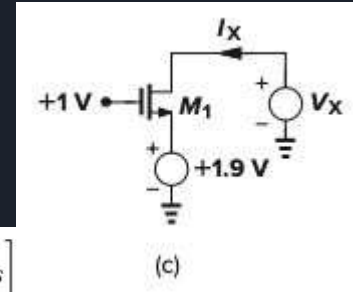
OBSERVATION

- Due to the symmetrical structure of the MOSFET, the effective source and drain terminals interchange when their relative voltages are reversed. Hence, for $V_X < 1V$, the source and drain are reversed, whereas for $V_X > 1V$, they assume their normal roles.

Problem 2.5-c

- 2.5. Sketch I_X and the transconductance of the transistor as a function of V_X for each circuit in Fig. 2.47 as V_X varies from 0 to V_{DD} . In part (a), assume that V_X varies from 0 to 1.5 V.

Diagram given -



Formula used -

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L'} (V_{GS} - V_{TH})^2$$

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

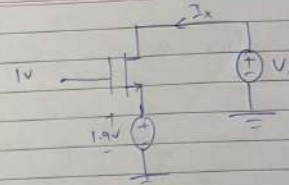
$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS} \text{ const.}} = \frac{2I_D}{V_{GS} - V_{TH}}$$

Handwritten Solution

Points which were noted -

- For $V_X < 1.9V$, the drain and source are interchanged.
- MOSFET is ON till $V_X = 0.3V$.
- For $V_X < 0.3V$, the MOSFET is in saturation

Given



h.v. $V_X = 1.9V$
S and D exchange their roles ($I_X \rightarrow -I_D$)

$V_{GS} = 1 - V_X$, $V_{DS} = 1.9 - V_X$

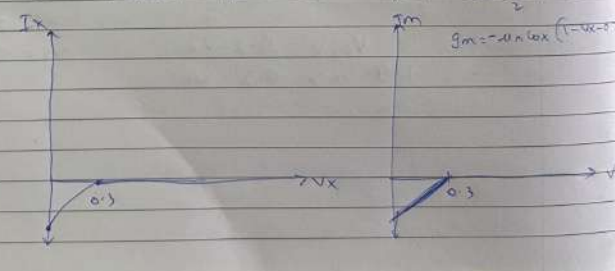
For MOSFET to turn ON
 $V_{GS} > V_{th}$

$1 - V_X > 0.7$
 $V_X < 0.3$

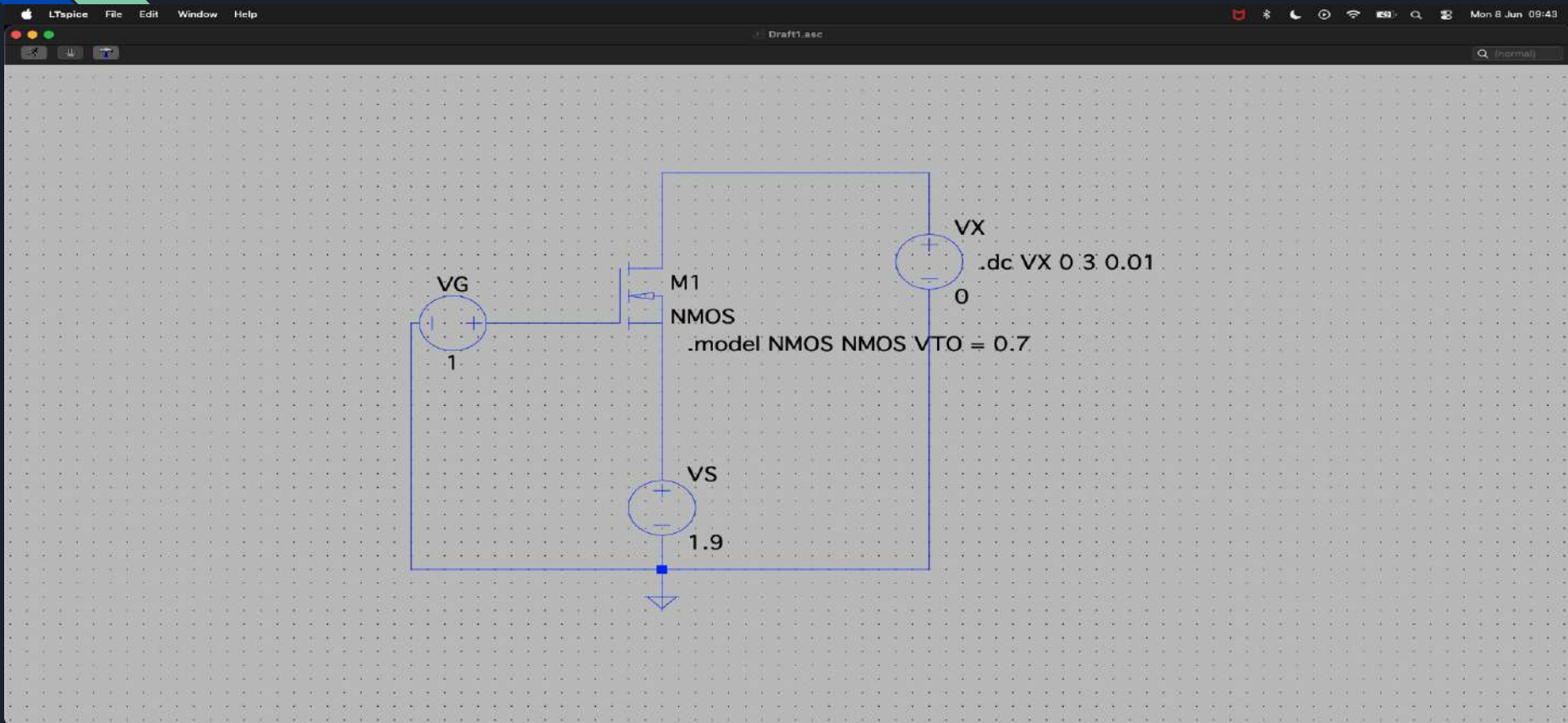
MOSFET turns OFF for $V_X > 0.3V$
for $V_X < 0.3$

$V_{DS} > V_{GS} - V_{th}$
 $1.9 - V_X > 1 - V_X - 0.7$

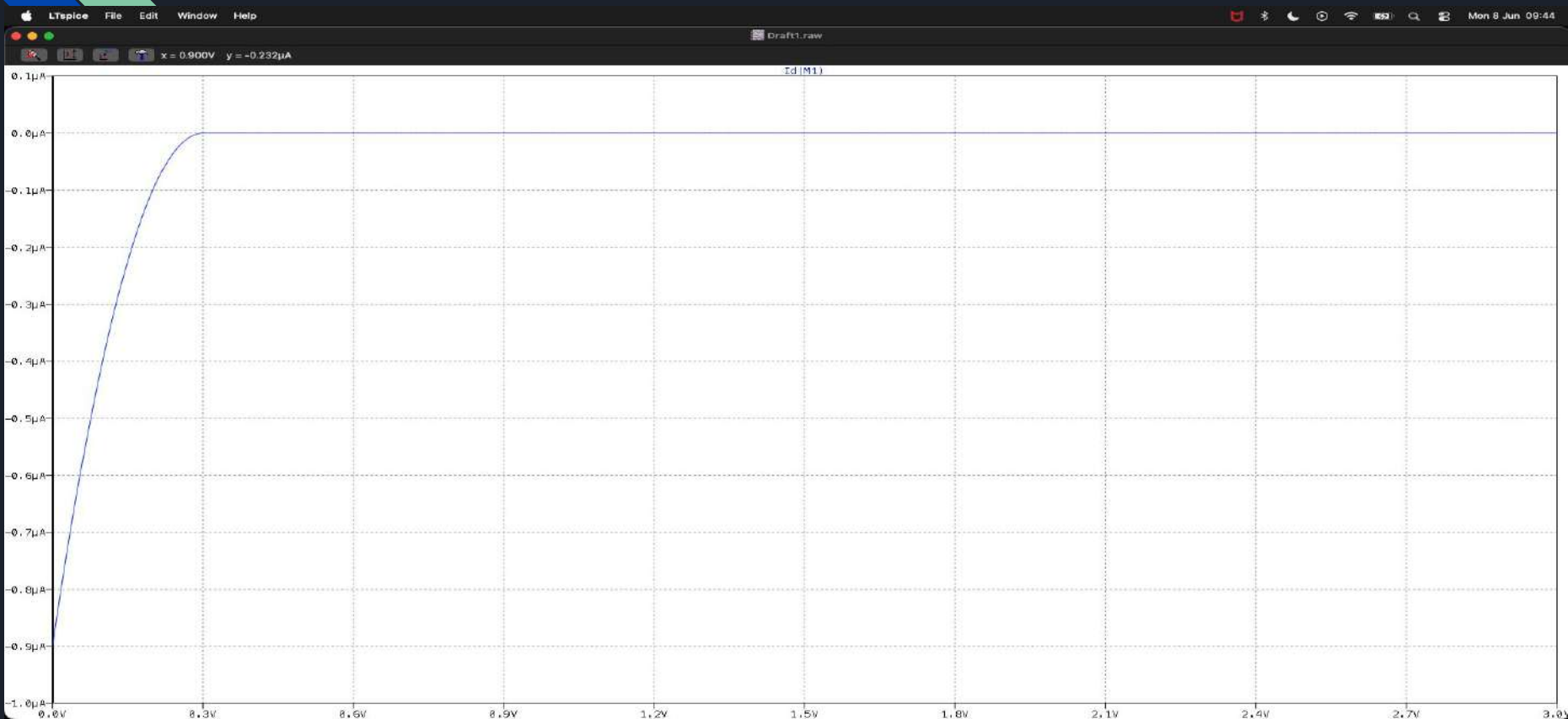
MOSFET is in saturation, $I_X = -I_D = -\frac{1}{2} \mu_n C_{ox} (W/L) (V_{GS} - V_{th})^2$
 $g_m = -\mu_n C_{ox} (W/L) (V_{GS} - V_{th})$



LTSPICE Schematic



Simulation Results for I_x





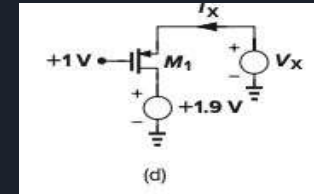
OBSERVATION

- Due to the symmetrical structure of the MOSFET, the effective source and drain terminals interchange when their relative voltages are reversed (direction of current reverses). Hence, for $V_X < 1.9\text{V}$, the source and drain are reversed, whereas for $V_X > 1.9\text{V}$, they assume their normal roles. But in this case the MOSFET turns OFF after $V_X = 0.3\text{V}$ therefore current becomes 0 for $V_X > 0.3\text{V}$.

Problem 2.5-d

- 2.5. Sketch I_X and the transconductance of the transistor as a function of V_X for each circuit in Fig. 2.47 as V_X varies from 0 to V_{DD} . In part (a), assume that V_X varies from 0 to 1.5 V.

Diagram given -



Formula used -

I_D = Saturation

$$I_D = -\frac{1}{2} \mu_p C_{ox} \frac{W}{L'} (V_{GS} - V_{TH})^2$$

Linear -

$$I_D = -\mu_p C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

g_m = Saturation

$$g_m = -k_p \frac{W}{L} (V_{GS} - V_{TP})$$

Linear -

$$g_m = -k_p \frac{W}{L} V_{DS}$$

Handwritten Solution

Points which were noted -

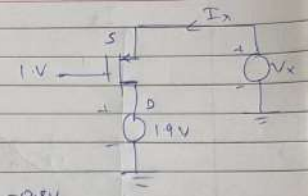
- For $V_X < 1.9V$, the drain and source are interchanged, current is reversed.
- For $V_X < 1.8V$ the MOSFET is in saturation, current is constant.
- For $V_X > 1.8$ the MOSFET is in linear region,

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$\therefore$ till $V_X = 1.9V$
D and S are interchanged.

$$V_{GS} = -0.9V$$

$$V_{DS} = V_X - 1.9V, \quad V_{th} = -0.8V$$



Case 1 $V_X < 1.8V$

$|V_{GS}| > |V_{th}| \rightarrow$ MOSFET turns ON
 $V_{GS} < V_{th}$

for saturation

$$V_{DS} < V_{GS} - V_{th}$$

$$V_X - 1.9 < -0.8$$

$$V_X < 1.9 \rightarrow \text{always true}$$

$\therefore$ MOSFET in saturation

$$I_X = -I_{DS} = -\frac{1}{2} \mu_{pox} \frac{W}{L} (-0.8)^2$$

$$g_m = \frac{\partial I_{DS}}{\partial V_{GS}} = -\mu_{pox} \frac{W}{L} (-0.8)$$

Case 2 $V_X \in (1.8, 1.9) \rightarrow$ linear region

$$I_X = -I_{DS} = -\mu_{pox} \frac{W}{L} \left[(-0.8)(V_X - 1.9) - \frac{1}{2}(V_X - 1.9)^2 \right]$$

$$g_m = -\mu_{pox} \frac{W}{L} (V_X - 1.9)$$

Case 3 $V_X > 1.9$ S and D interchange again

$$V_{GS} = 1 - V_X, \quad V_{th} = 1.8 - V_X$$

$$V_{GS} - V_{th} = (1 - V_X) - (1.8 - V_X)$$

$$= 1.8 - V_X$$

$$V_{DS} > V_{GS} - V_{th} \rightarrow \text{linear}$$

$$\therefore 1.9 - V_X > 1.8 - V_X$$

$$1.9 > 1.8 \rightarrow \text{always true}$$

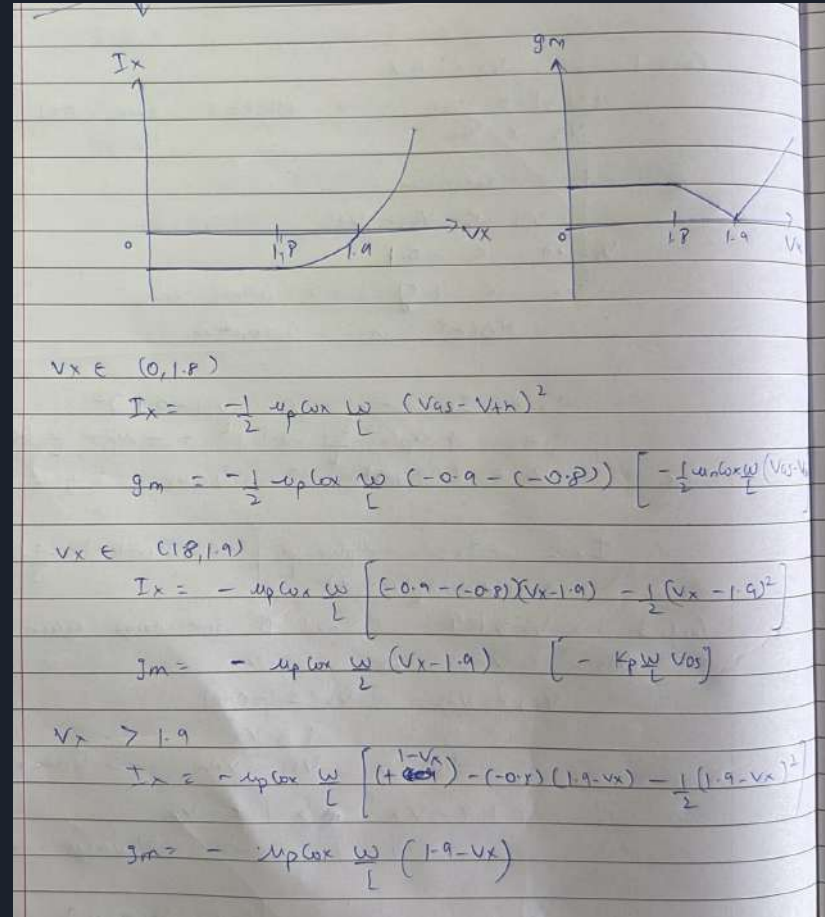
$$I_X = I_{DS} = \frac{1}{2} \mu_{pox} \frac{W}{L} \left[(1.8 - V_X)(1.9 - V_X) - \frac{1}{2}(1.9 - V_X)^2 \right]$$

$$g_m = \mu_{pox} \frac{W}{L} (1.9 - V_X)$$

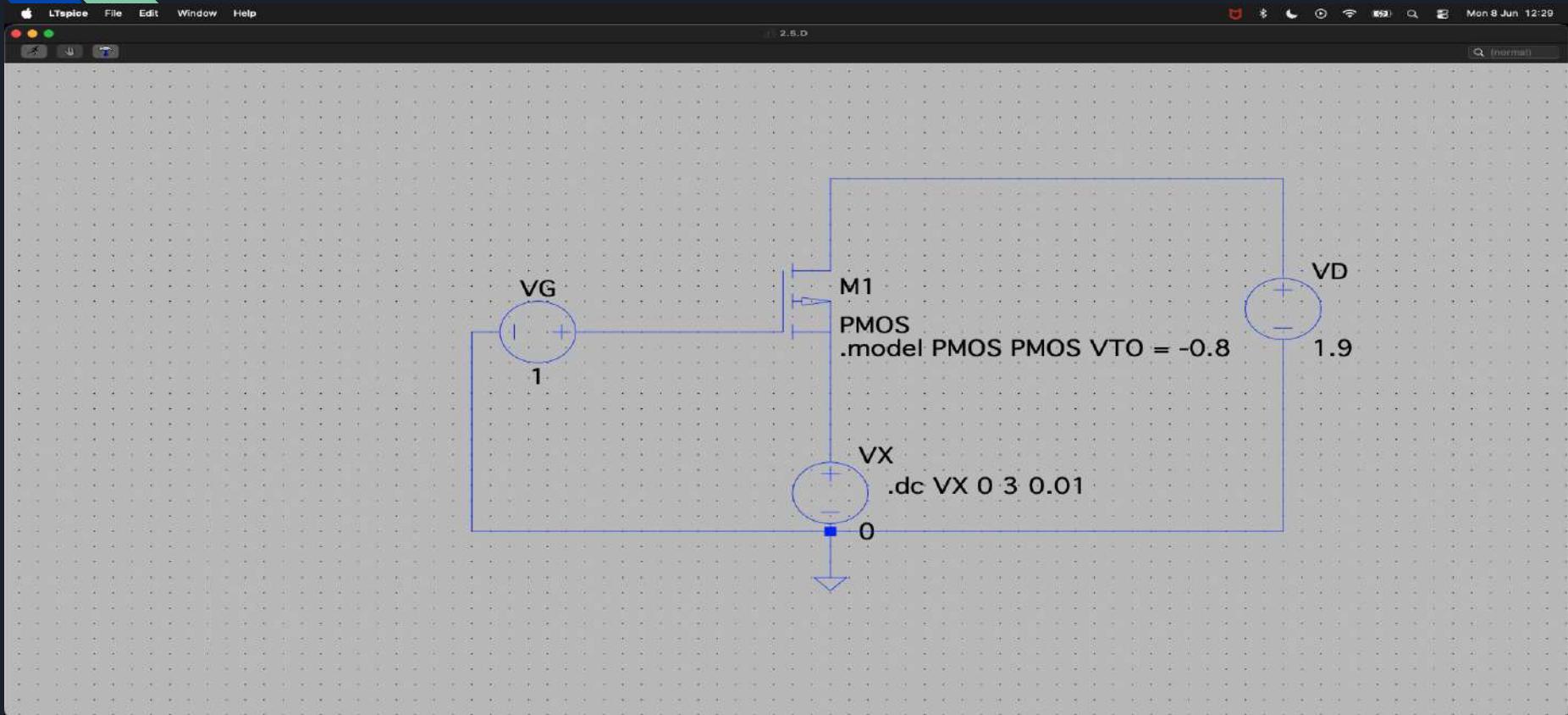
Handwritten Solution

Points which were noted -

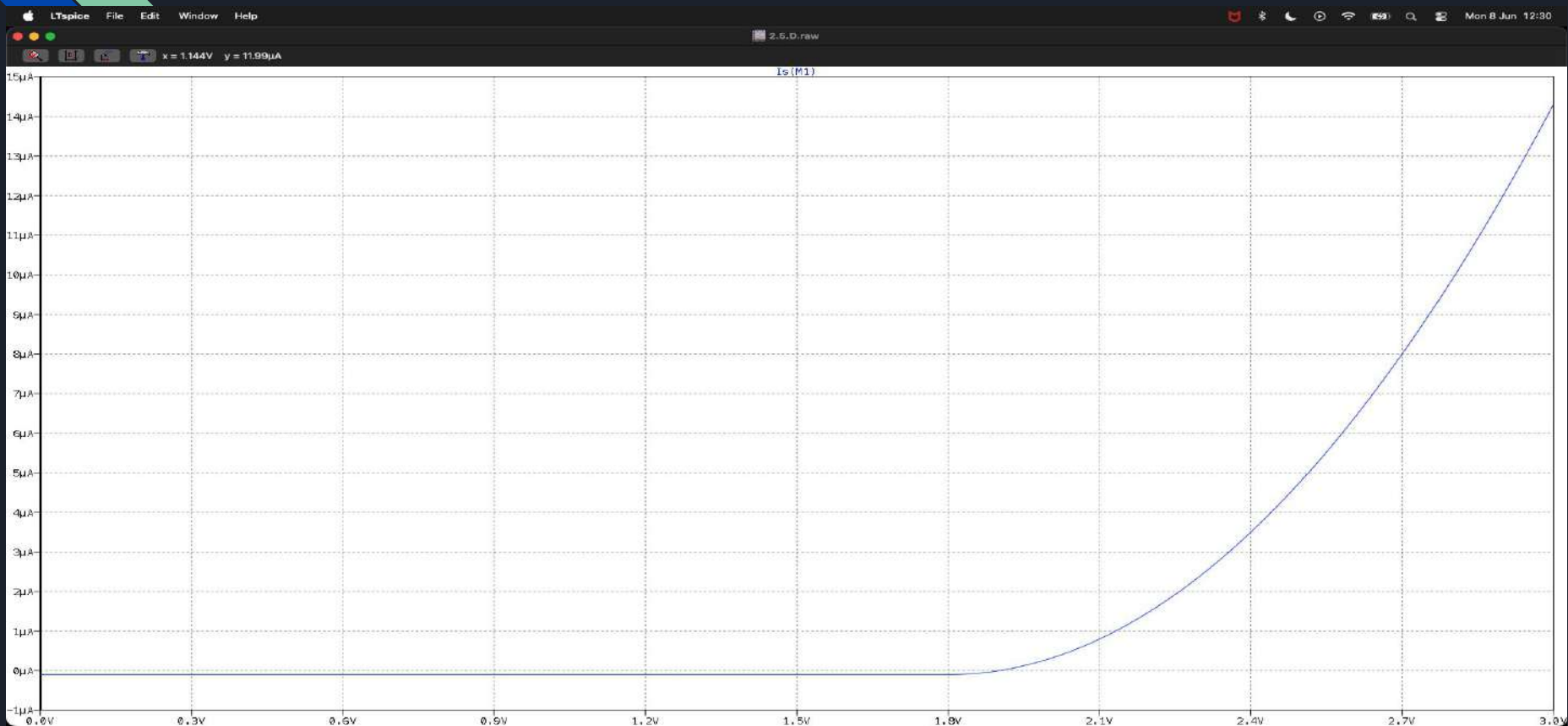
- For $V_X < 1.9V$, the drain and source are interchanged.
- For $V_X < 1.8V$ the MOSFET is in saturation.
- For $V_X > 1.8$ the MOSFET is in linear region



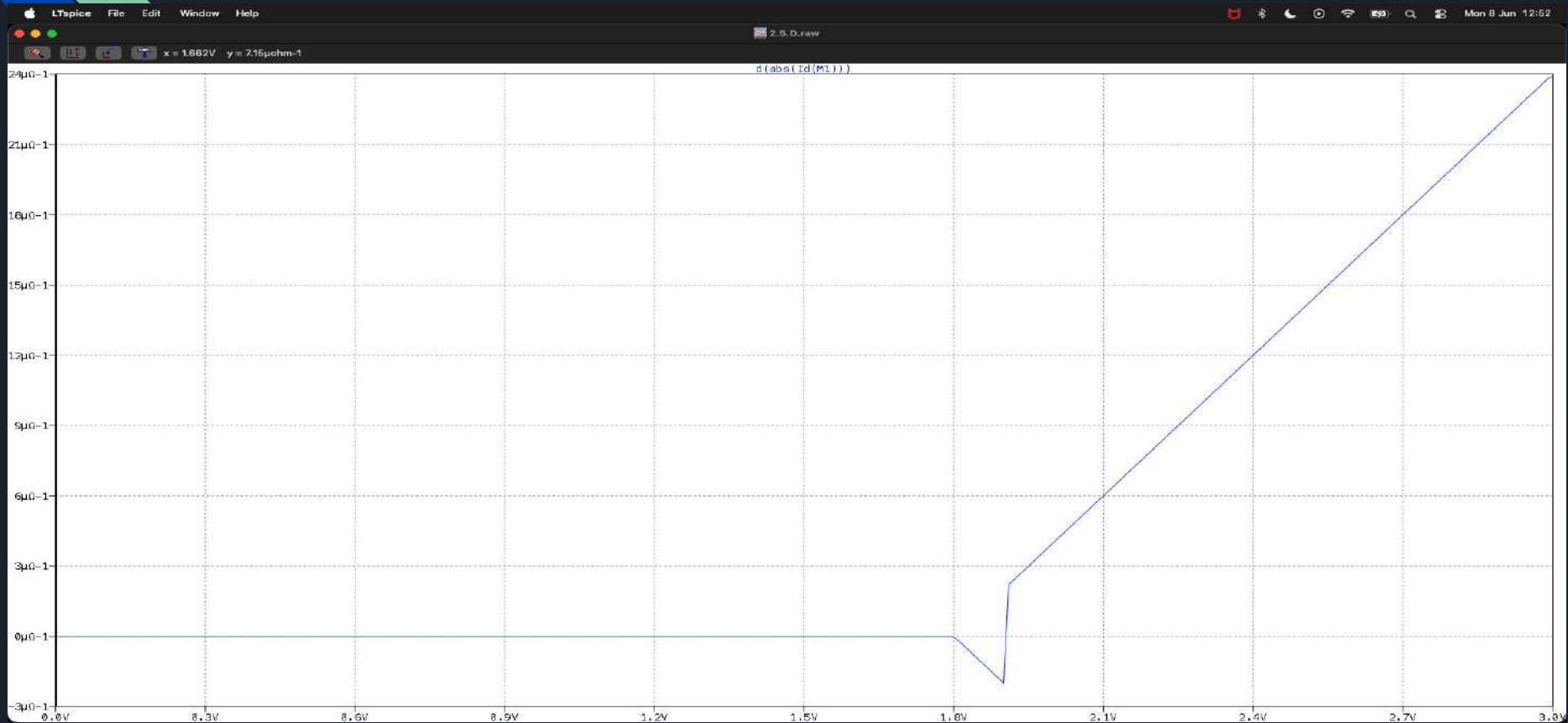
LTSPICE Schematic



Simulation Results for Ix



Simulation Results for Ix





OBSERVATION

- Due to the symmetric structure of the MOSFET, the effective source and drain terminals interchange when their voltage polarity is reversed, causing the direction of current flow to reverse. Therefore, for $V_X < 1.9\text{V}$, the source and drain effectively swap roles, while for $V_X > 1.9\text{V}$, they operate in their normal configuration.
- In this region, the drain current is very small because the overdrive voltage $V_{GS} - V_{TH}$ is only 0.1V . As a result, the saturation current I_{Dsat} is approximately zero.
- When V_X exceeds 1.9V , the MOSFET enters the linear (triode) region. As V_X increases further, the channel becomes more conductive, causing the drain current to increase.

Problem 2.5-e

- 2.5. Sketch I_X and the transconductance of the transistor as a function of V_X for each circuit in Fig. 2.47 as V_X varies from 0 to V_{DD} . In part (a), assume that V_X varies from 0 to 1.5 V.

Diagram given -

Formula used -

I_D = Saturation

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

Linear -

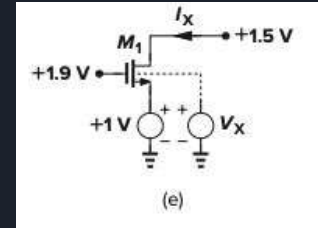
$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

g_m = Saturation

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})$$

Linear -

$$g_m = \mu_n C_{ox} \frac{W}{L} V_{DS}$$



Handwritten Solution

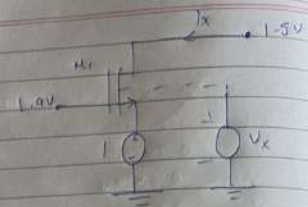
Points Noted:

- The body-effect equation is derived assuming a long-channel MOSFET and a source-body junction that is not significantly forward biased (depletion-region approximation-Shichman-Hodges model).
- Therefore, V_X is considered only up to 1.9V, which is the limit of validity of the body-effect/Shichman-Hodges analysis used in this calculation.
- For higher values of V_X , the source-body junction becomes increasingly forward biased, violating the assumptions of the body-effect and Shichman-Hodges models, and the resulting equations no longer accurately describe the MOSFET behavior.

2.5V_{DD}

Given

$V_{GS} = 0.9V$
 $V_{DS} = 0.5V$
 $V_{SB} = 1 - V_X$



$V_{th} = V_{th0} + \gamma \left[\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right]$

$\gamma = 0.45$, $\phi_F = 0.9$, $V_{th0} = 0.7V$

$V_{th} = 0.7 + 0.45 \left[\sqrt{0.9 + 1 - V_X} - \sqrt{0.9} \right]$

for $V_X = 0$, $V_{th} = 0.813V$, MOSFET is in saturation

$I_X = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left(0.9 - 0.7 - 0.45 \left(\sqrt{0.9 + V_X} - \sqrt{0.9} \right) \right)^2$

$g_m = \mu_n C_{ox} \frac{W}{L} \left(0.9 - 0.7 - 0.45 \left(\sqrt{0.9 + V_X} - \sqrt{0.9} \right) \right)$

this is valid till MOSFET enters triode region,

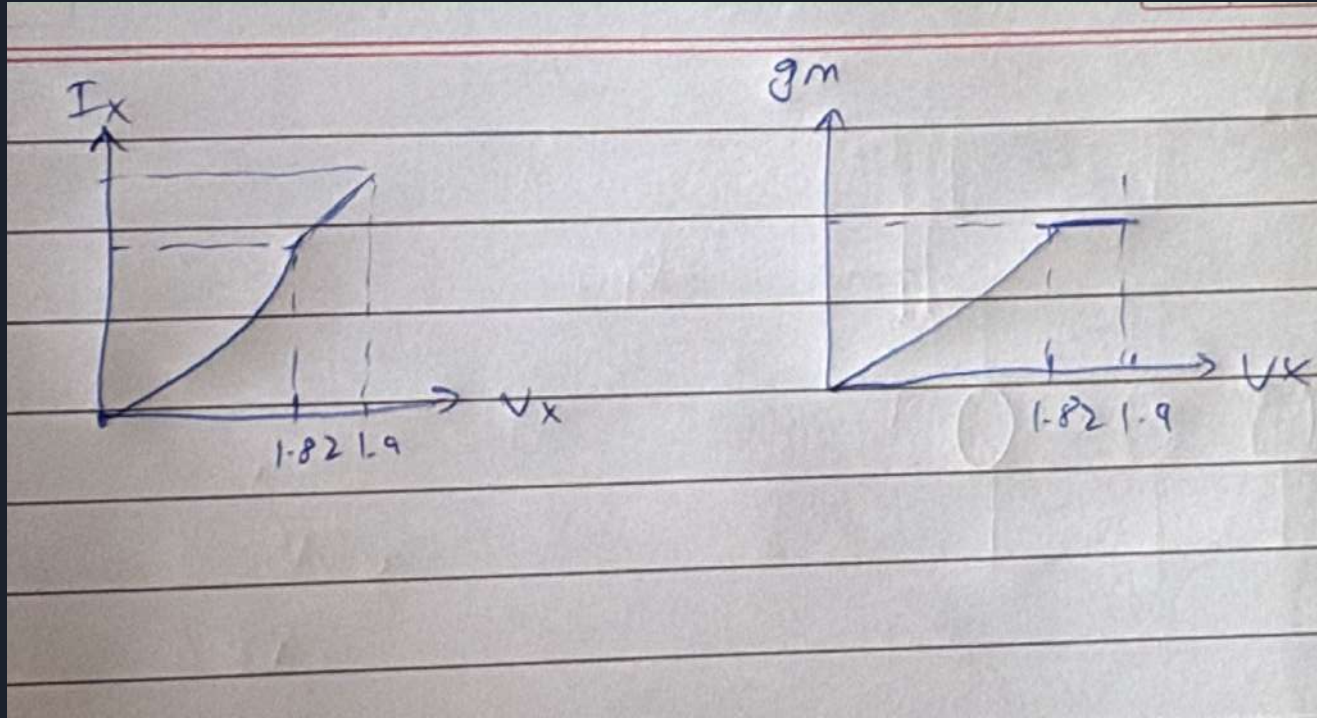
for triode region

$V_{DS} < V_{GS} - V_{th}$
 $0.5 < 0.9 - V_{th}$
 $V_{th} < 0.4$, $V_X > 1.82V$

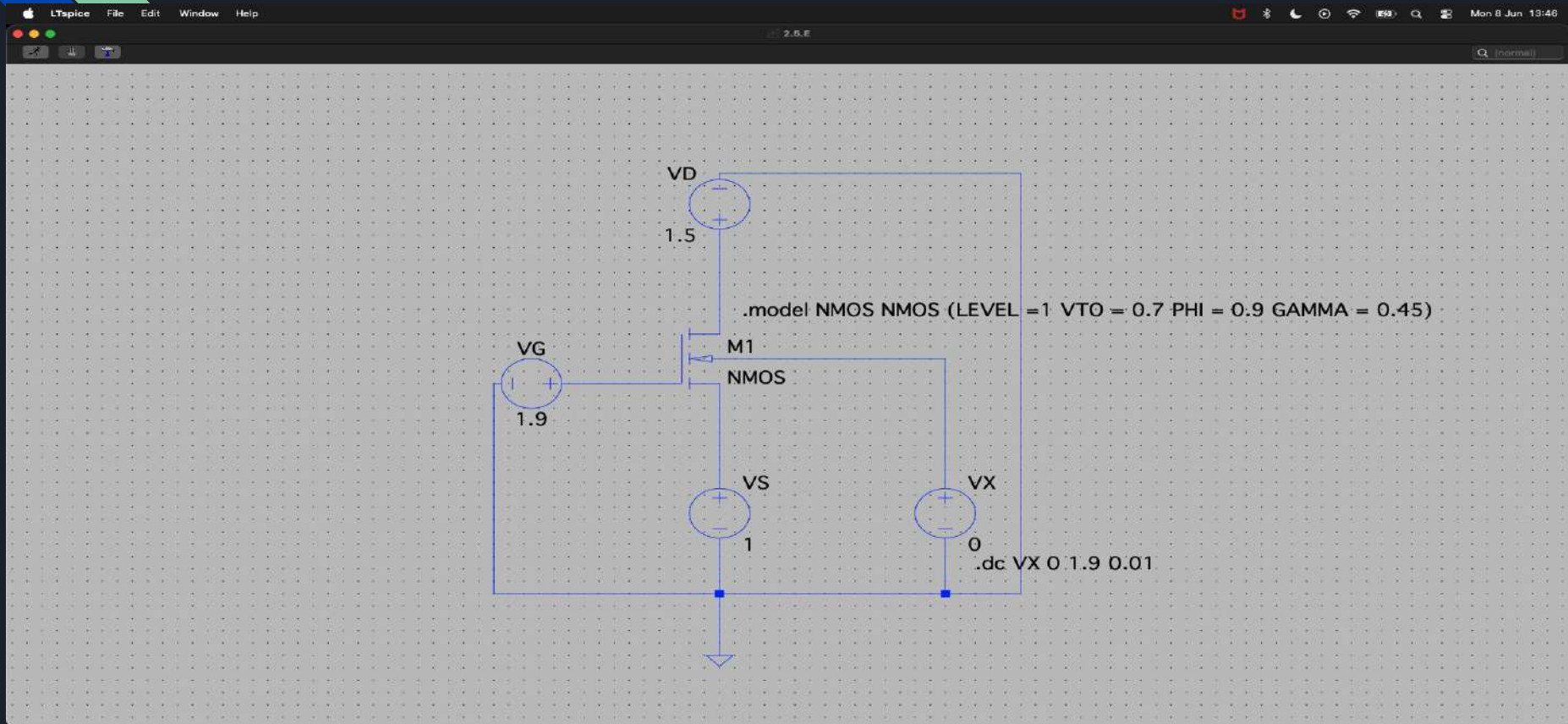
$V_X > 1.82V \rightarrow$ linear region

We only consider V_X till 1.9V as after that the Shichman-Hodges eqn is not valid.

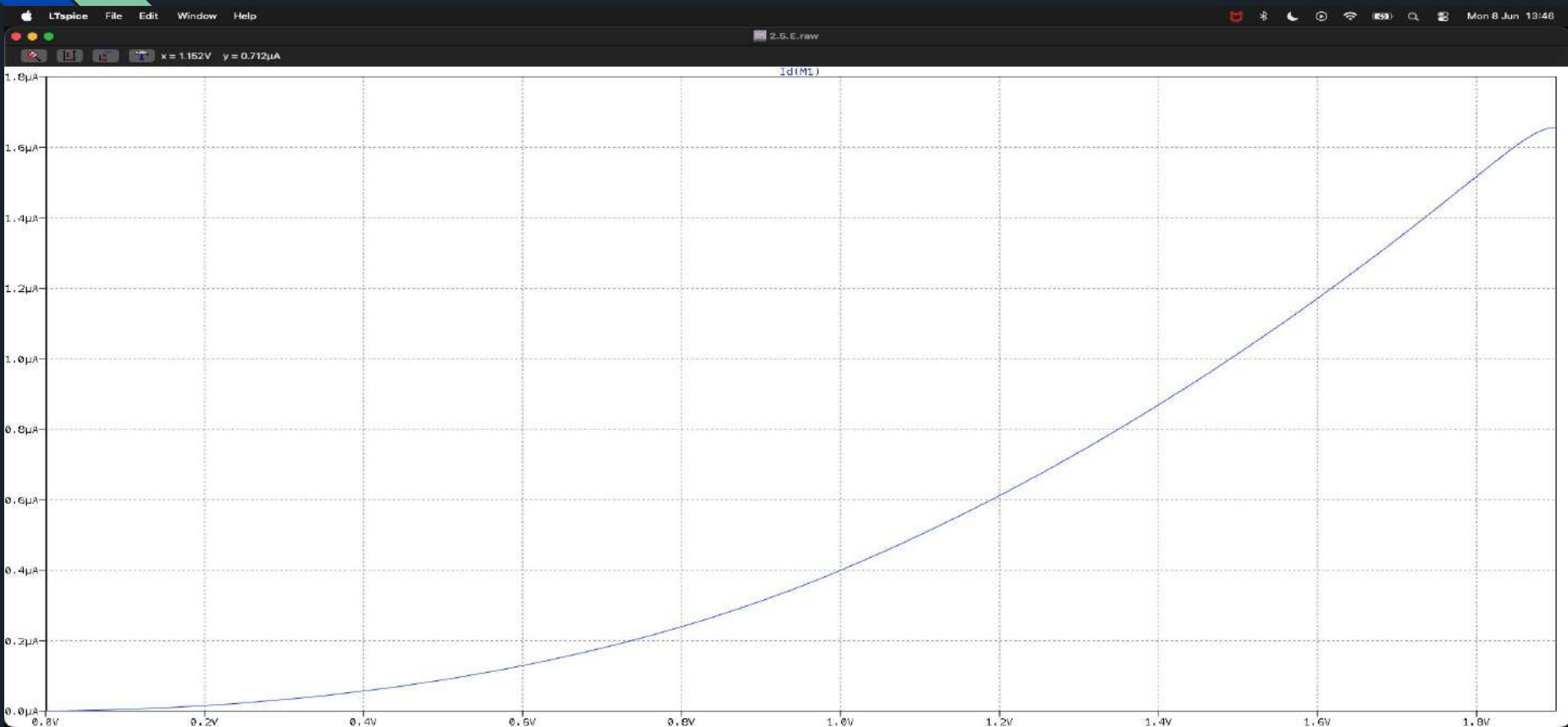
Handwritten Solution



LTSPICE Schematic



Simulation Results for I_x

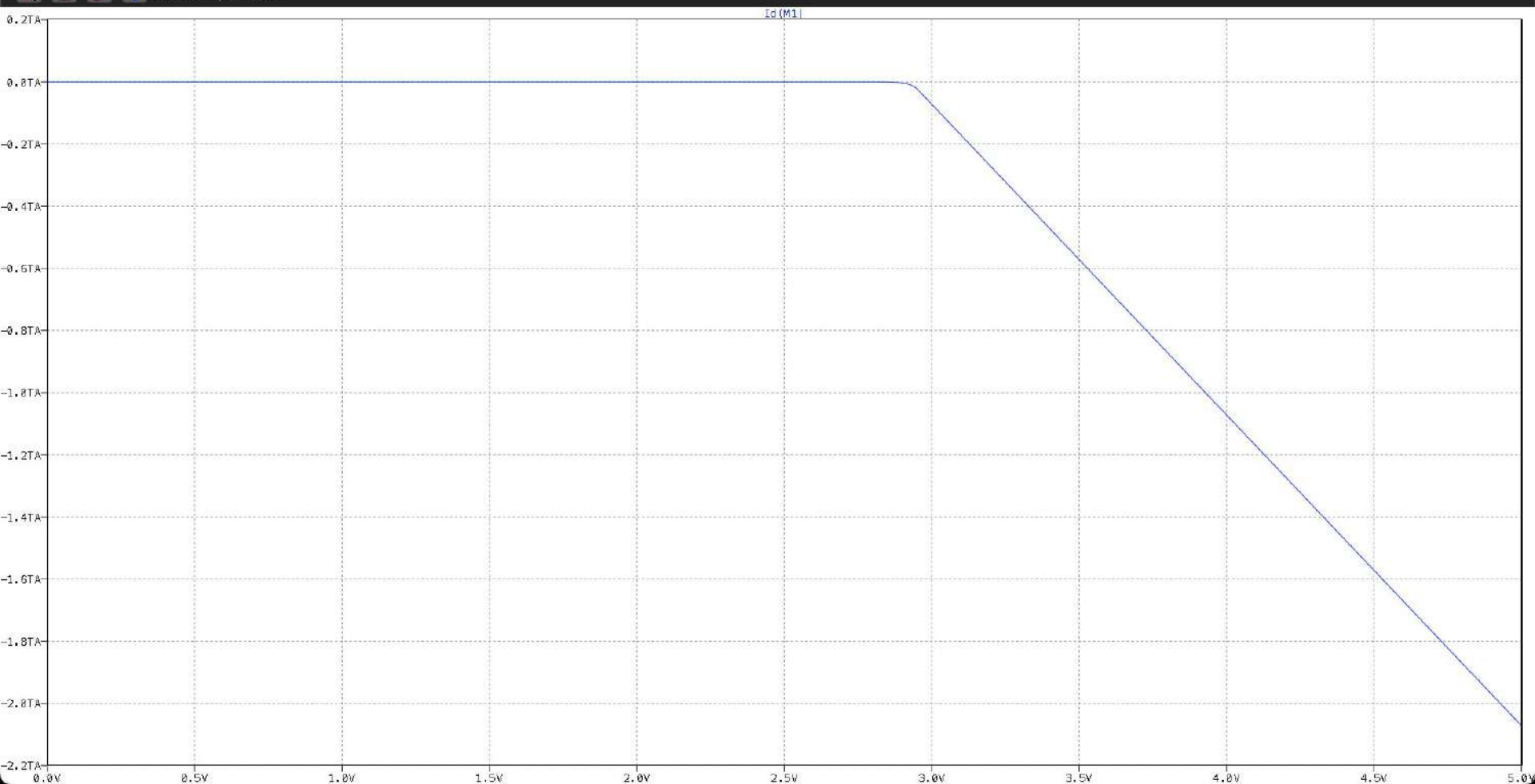




What will happen if V_x exceeds 1.9V?

When V_X is swept beyond the valid range of the body-effect/Shichman-Hodges model, the source-body junction becomes strongly forward biased, leading to excessive parasitic diode conduction. As a result, the simulator predicts extremely large currents that are not physically meaningful. The negative sign is due to the current reference direction used by LTspice, while the tera-amp magnitude indicates that the device model is being operated outside its region of validity.

x = 2.988V y = -1.030TA





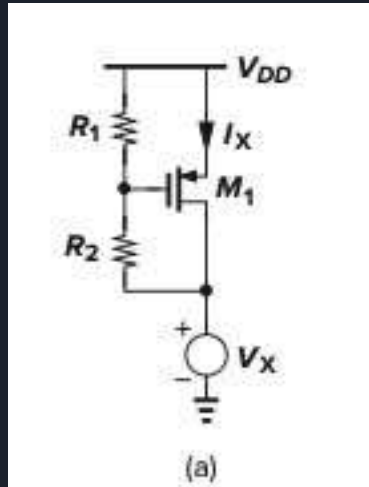
OBSERVATION

- The sweep of V_X is restricted to $V_X < 1.9V$ because the analysis is based on the Shichman–Hodges MOSFET model together with the body-effect equation for threshold voltage. The source-to-bulk voltage is given by $V_{SB} = 1 - V_X$). Therefore, the source-body junction becomes forward biased when $V_X > 1V$. Although the forward bias is initially small, increasing V_X further violates the depletion-region assumptions used in the derivation of the body-effect model.
- At $V_X = 1.9V$, the argument of the body-effect term reaches its limiting value: $2\phi_F + V_{SB} = 0$.
- Beyond this point, the assumptions underlying the body-effect and Shichman–Hodges models are no longer valid, as the source-body junction is significantly forward biased and substantial parasitic diode conduction can occur. Consequently, the threshold-voltage and drain-current equations no longer accurately describe the MOSFET behavior.
- Therefore, only the range $V_X < 1.9V$ is considered for theoretical analysis and comparison with simulation results.

Problem 2.6-a

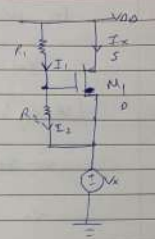
2.6. Sketch I_X and the transconductance of the transistor as a function of V_X for each circuit in Fig. 2.48 as V_X varies from 0 to V_{DD} .

Diagram -



Handwritten Solution

26xV



$V_{gs} = V_g - V_s = V_g - V_{DD}$
 $V_{ds} = V_d - V_s = -V_{DD} + V_x$

$\therefore$ we know $I_d = 0$
 $I_1 = I_2$

$\frac{V_{DD} - V_g}{R_1} = \frac{V_g - V_x}{R_2}$

$\frac{V_{DD} - V_g}{R_1} = \frac{V_g \left(\frac{1 + R_1}{R_1} \right)}{R_2}$

$\frac{V_{DD} R_2 + V_x R_1}{R_1 R_2} = \frac{V_g R_1 + R_2}{R_1 R_2}$

$V_g = \frac{V_{DD} R_2 + V_x R_1}{R_1 + R_2}$

$\therefore V_{gs} = V_g - V_{DD} = \frac{V_{DD} R_2 + V_x R_1}{R_1 + R_2} - V_{DD}$

$= \frac{V_x R_1}{R_1 + R_2} + V_{DD} \left(\frac{R_2}{R_1 + R_2} - 1 \right)$

$= \frac{V_x R_1}{R_1 + R_2} + \frac{V_{DD} - R_1}{R_1 + R_2}$

$V_{gs} = \left(\frac{R_1}{R_1 + R_2} \right) (V_x - V_{DD})$

For MOSFET to turn ON

$V_{gs} < V_{tp}$

$\left(\frac{V_x - V_{DD}}{R_1 + R_2} \right) R_1 < V_{tp}$

$V_x - V_{DD} < \frac{V_{tp} R_1 + V_{tp} R_2}{R_1}$

$V_x < V_{tp} + \frac{V_{tp} R_2}{R_1} + V_{DD}$

$V_x < V_{tp} \left(1 + \frac{R_2}{R_1} \right) + V_{DD} - \text{MOSFET OFF after this}$

for saturation

$V_{gs} - V_{tp} > V_{ds}$

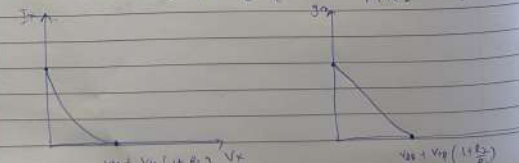
$\left(\frac{V_x - V_{DD}}{R_1 + R_2} \right) R_1 - V_{tp} > (V_x - V_{DD})$

$\therefore$ This is always true
 Device is saturation if it is in ON

$g_m = -\mu_p C_{ox} \frac{W}{L} (V_{gs} - V_{tp}) = -\mu_p C_{ox} \frac{W}{L} \left(\frac{V_x - V_{DD}}{R_1 + R_2} R_1 - V_{tp} \right)$

$I_d = I_s = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{gs} - V_{tp})^2$

$I_d = I_s = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} \left(\frac{V_{DD} + V_x}{R_1 + R_2} R_1 - V_{tp} \right)^2$

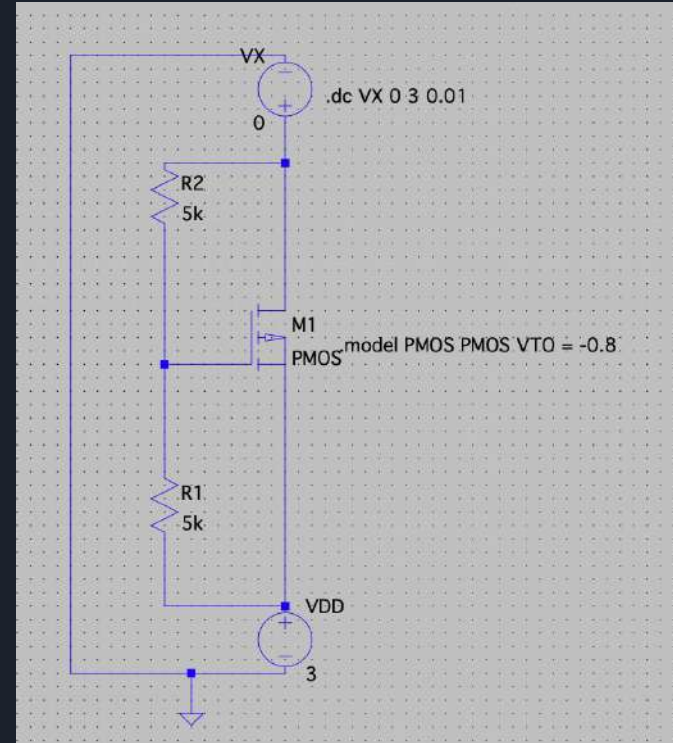


LTSPICE Schematic

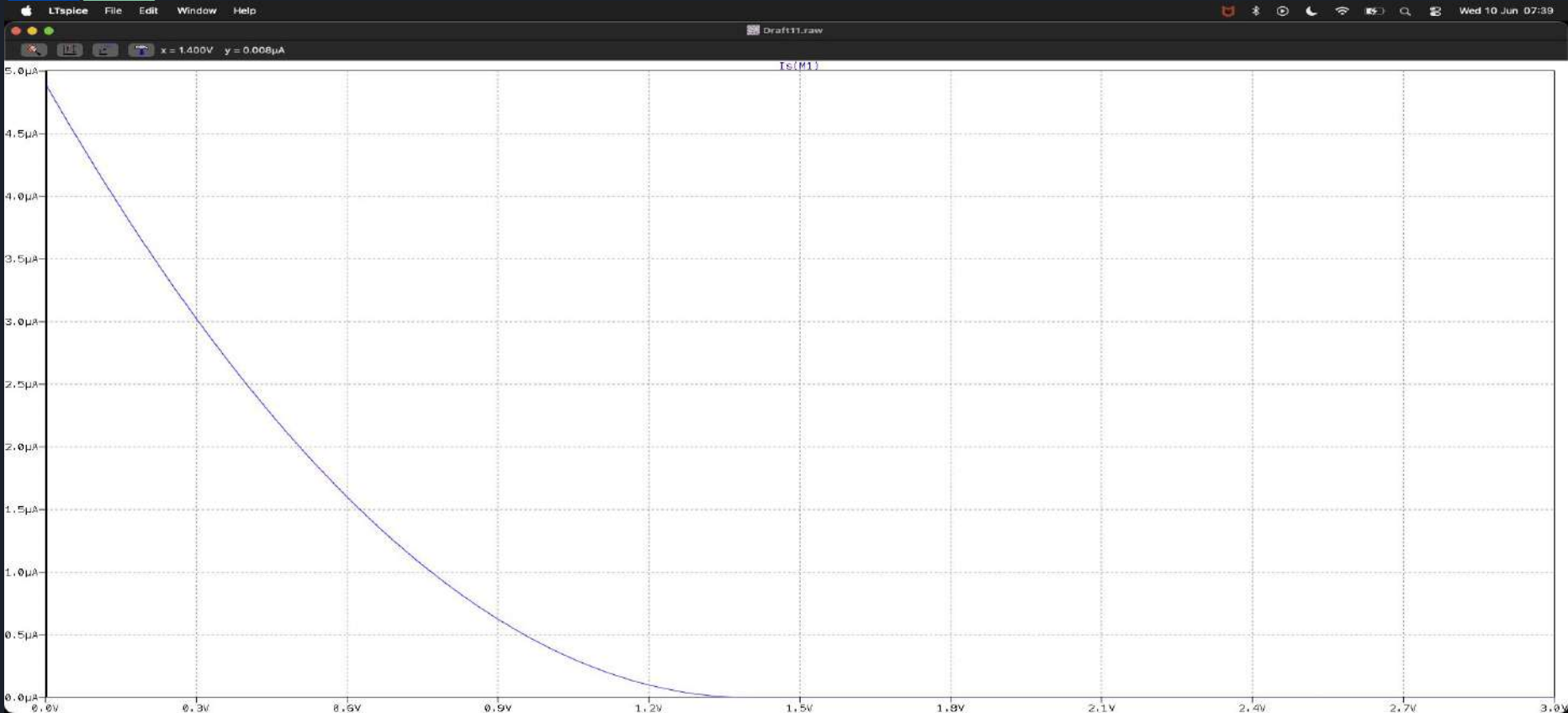
PAGE NO.

Let us Assume $R_1 = R_2 = 5k\Omega$, $V_{DD} = 3V$, $V_{tp} = -0.8V$

edge voltage = $V_{DD} + V_{tp} \left(\frac{1+R_2}{R_1} \right)$

$$\begin{aligned} &= 3 - 0.8 (1+1) \\ &= 3 - 0.8(2) \\ &= 3 - 1.6 \\ &= 1.4V \end{aligned}$$


Simulation Results for I_x



Simulation Results for gm





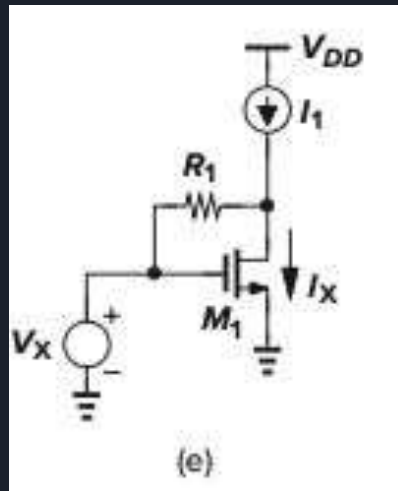
OBSERVATION

- R1 and R2 form a voltage-divider bias network that sets the MOSFET gate voltage and hence its operating point. By controlling V_{GS} , they determine whether the device operates in cutoff, triode, or saturation region.

Problem 2.6-e

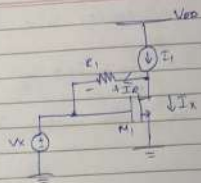
2.6. Sketch I_X and the transconductance of the transistor as a function of V_X for each circuit in Fig. 2.48 as V_X varies from 0 to V_{DD} .

Diagram -



Handwritten Solution

24/04



$$V_{gs} = V_x$$

$$V_{ds} = V_D$$

$$V_x + I_D R_i = V_D$$

$$I_D = I_1 - I_x$$

$$V_D = V_x + R_i (I_1 - I_x)$$

when $V_{gs} = V_x < V_{th}$, MOSFET is OFF

$$I_D = 0, g_m = 0$$

Once V_x crosses V_{th} , MOSFET enters saturation region

$$V_{gs} > V_{gs} - V_{th}$$

$$V_x + R_i (I_1 - I_x) > V_x - V_{th}$$

$$R_i (I_1 - I_x) > -V_{th}$$

$$\therefore \text{initially } I_x = 0$$

MOSFET enters saturation

$$I_x = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_x - V_{th})^2$$

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_x - V_{th})$$

MOSFET will enter triode region when $V_{ds} < V_{gs} - V_{th}$. Let us consider equality to find transition point

$$V_{ds} = V_{gs} - V_{th}$$

$$V_x + R_i (I_1 - I_x) = V_x - V_{th}$$

$$I_x = \frac{V_{th}}{R_i} + I_1$$

$$I_x = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_x - V_{th})^2$$

$$\frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_x - V_{th})^2 = I_1 + \frac{V_{th}}{R_i}$$

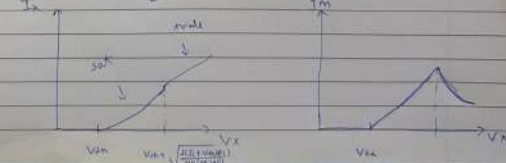
$$V_x = V_{th} + \sqrt{\frac{2(I_1 + V_{th}/R_i)}{\mu_n C_{ox} W/L}}$$

$$V_{th} < V_x < V_{th} + \sqrt{\frac{2(I_1 + V_{th}/R_i)}{\mu_n C_{ox} W/L}} \rightarrow \text{saturation}$$

$$V_x > V_{th} + \sqrt{\frac{2(I_1 + V_{th}/R_i)}{\mu_n C_{ox} W/L}} \rightarrow \text{triode}$$

$$I_x = \mu_n C_{ox} \frac{W}{L} (V_x - V_{th}) (V_x + V_{th} + R_i (I_1 - I_x)) - \frac{1}{2} (V_x + R_i (I_1 - I_x))^2$$

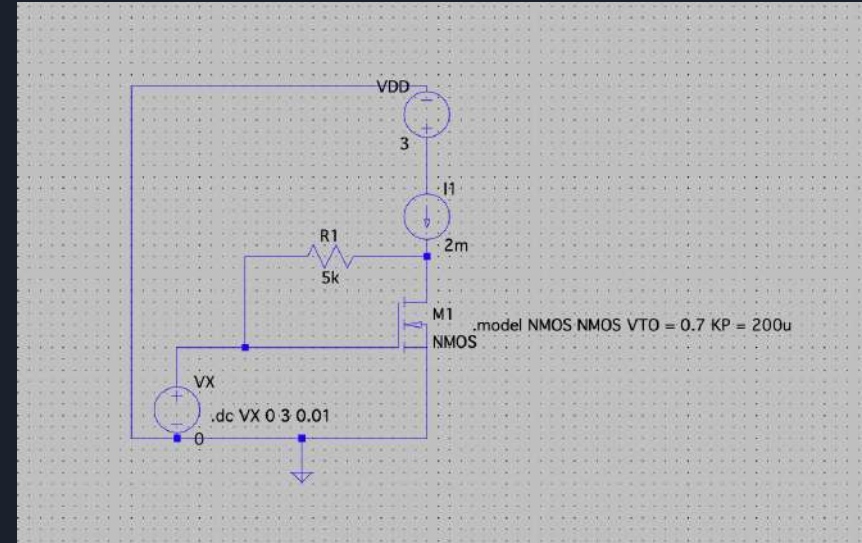
$$g_m = \mu_n C_{ox} \frac{W}{L} (V_x + R_i (I_1 - I_x))$$



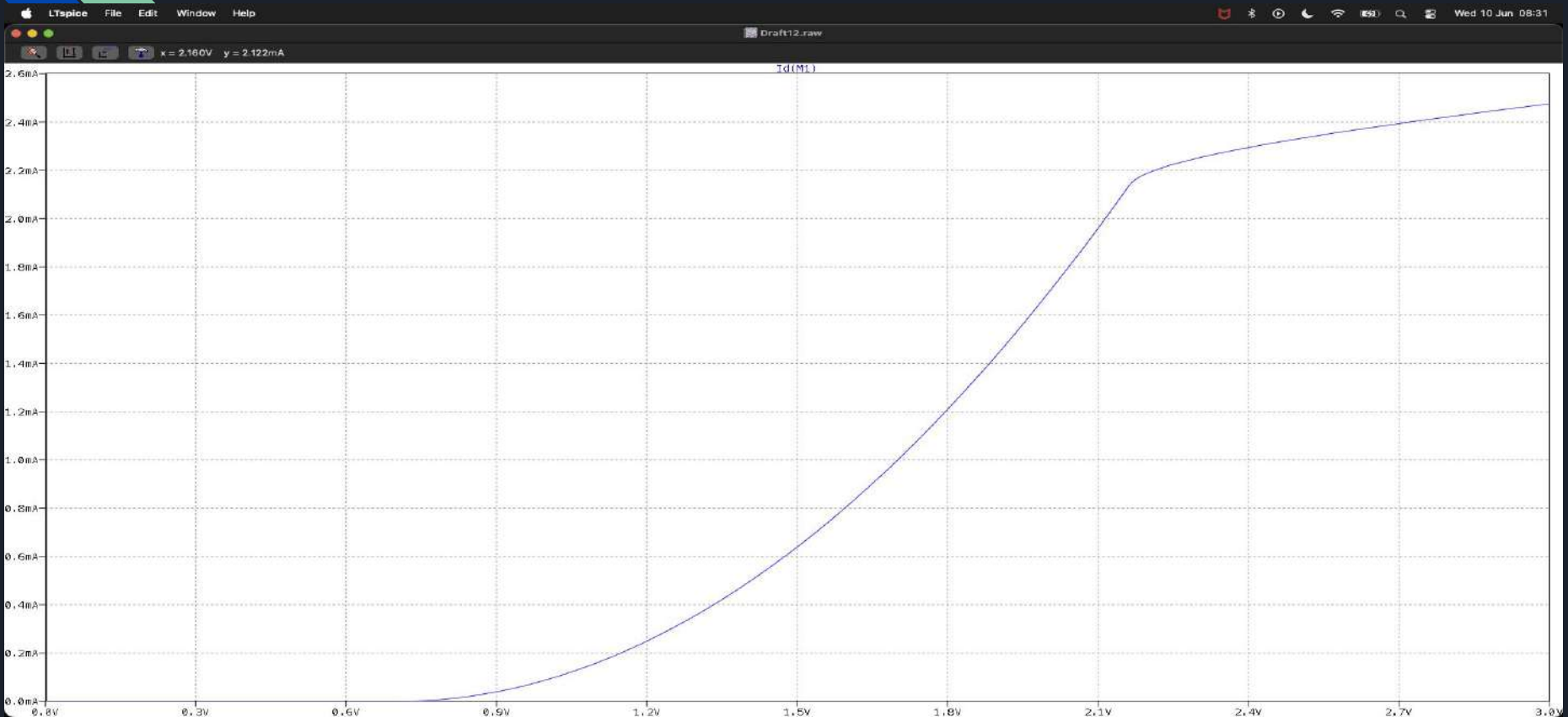
LTSPICE Schematic

Let us assume $I_1 = 2 \text{ mA}$, $R_1 = 5 \text{ k}$
 $\mu_n C_{ox} = 200 \mu\text{A/V}^2$, $W = 10 \mu\text{m}$, $L = 1 \mu\text{m}$

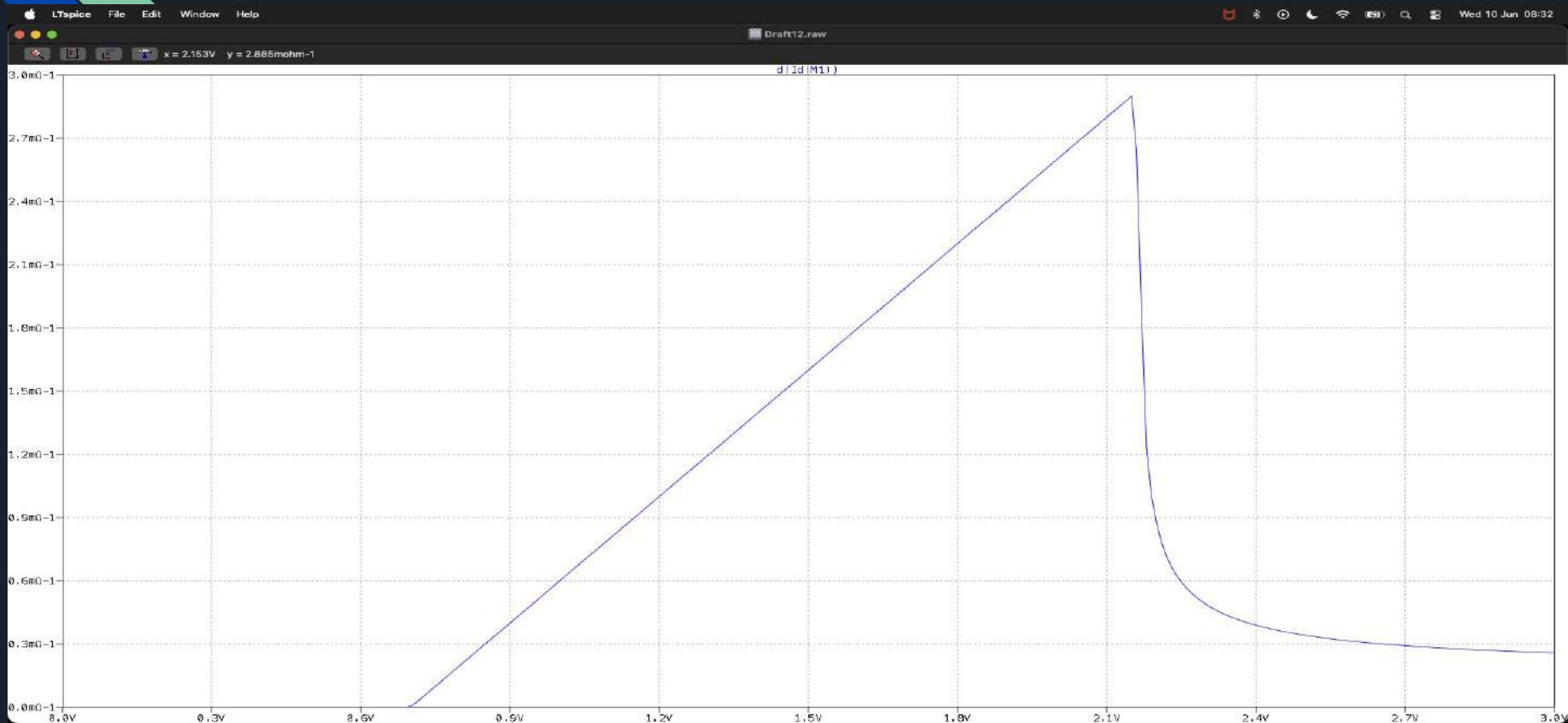
$$\text{threshold voltage} = V_{th} + \frac{\sqrt{2(I_1 + V_{th}/R_1)}}{\mu_n C_{ox}(W/L)}$$
$$= 2.16 \text{ V}$$



Simulation Results for I_x



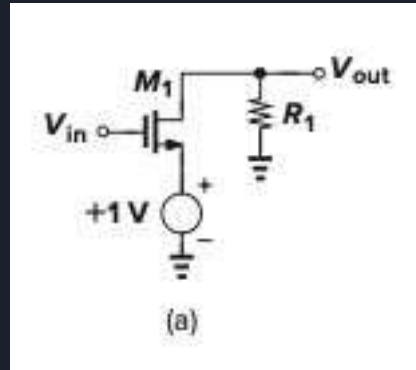
Simulation Results for gm



Problem 2.7-a

2.7. Sketch V_{out} as a function of V_{in} for each circuit in Fig. 2.49 as V_{in} varies from 0 to V_{DD} .

Diagram -



Handwritten Solution

2.7.2019

The drain and source get interchanged

$V_{GS} = V_{in} - V_{out}$
 $V_{DS} = 1 - V_{out}$

Initially $V_{out} = 0$, $\therefore$ if $V_{in} < V_{th} = 0.7V$
 MOSFET is OFF

once V_{in} crosses V_{th}

$V_{DS} > V_{GS} - V_{th}$
 $1 - V_{out} > V_{in} - V_{out} - 0.7$

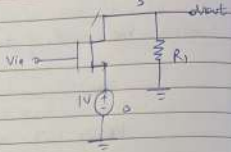
$V_{in} < 1.7V \rightarrow$ saturation

$\therefore$ $0 < V_{in} < 0.7 \rightarrow$ OFF
 $0.7 < V_{in} < 1.7 \rightarrow$ SATURATION
 $V_{in} > 1.7 \rightarrow$ TRIODE

for $V_{in} \in (0.7, 1.7)$
 $I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{out} - V_{th})^2$

for $V_{in} \in (1.7, 5)$
 $I_D = \mu_n C_{ox} \frac{W}{L} [(V_{in} - V_{out} - V_{th})(1 - V_{out}) - \frac{(1 - V_{out})^2}{2}]$

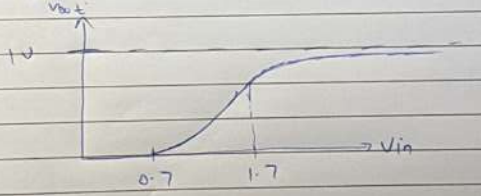
$V_{out} = I_D R_1 = R_1 \mu_n C_{ox} \frac{W}{L} [(V_{in} - V_{out} - V_{th})(1 - V_{out}) - \frac{(1 - V_{out})^2}{2}]$



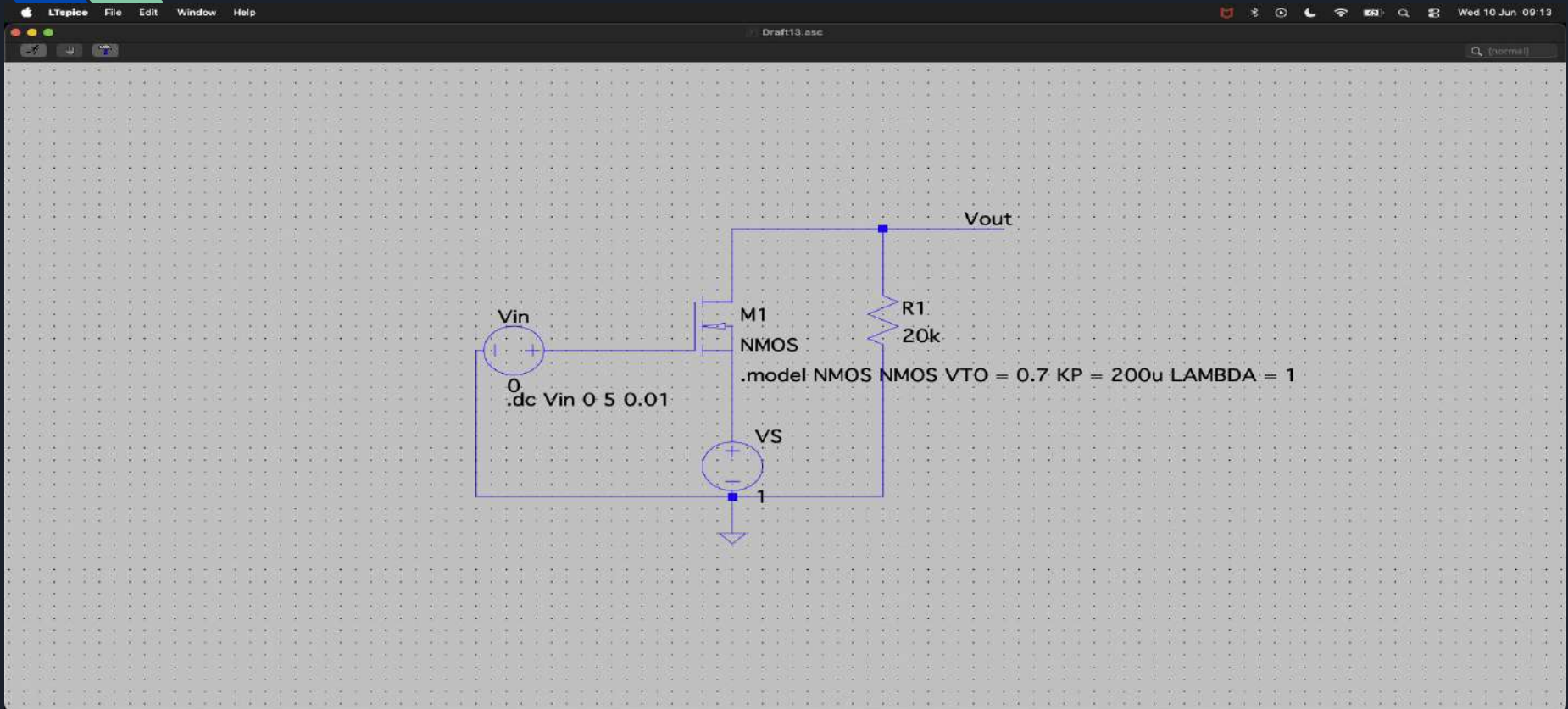
equivalent ckt

V_{out} will tend to 1V

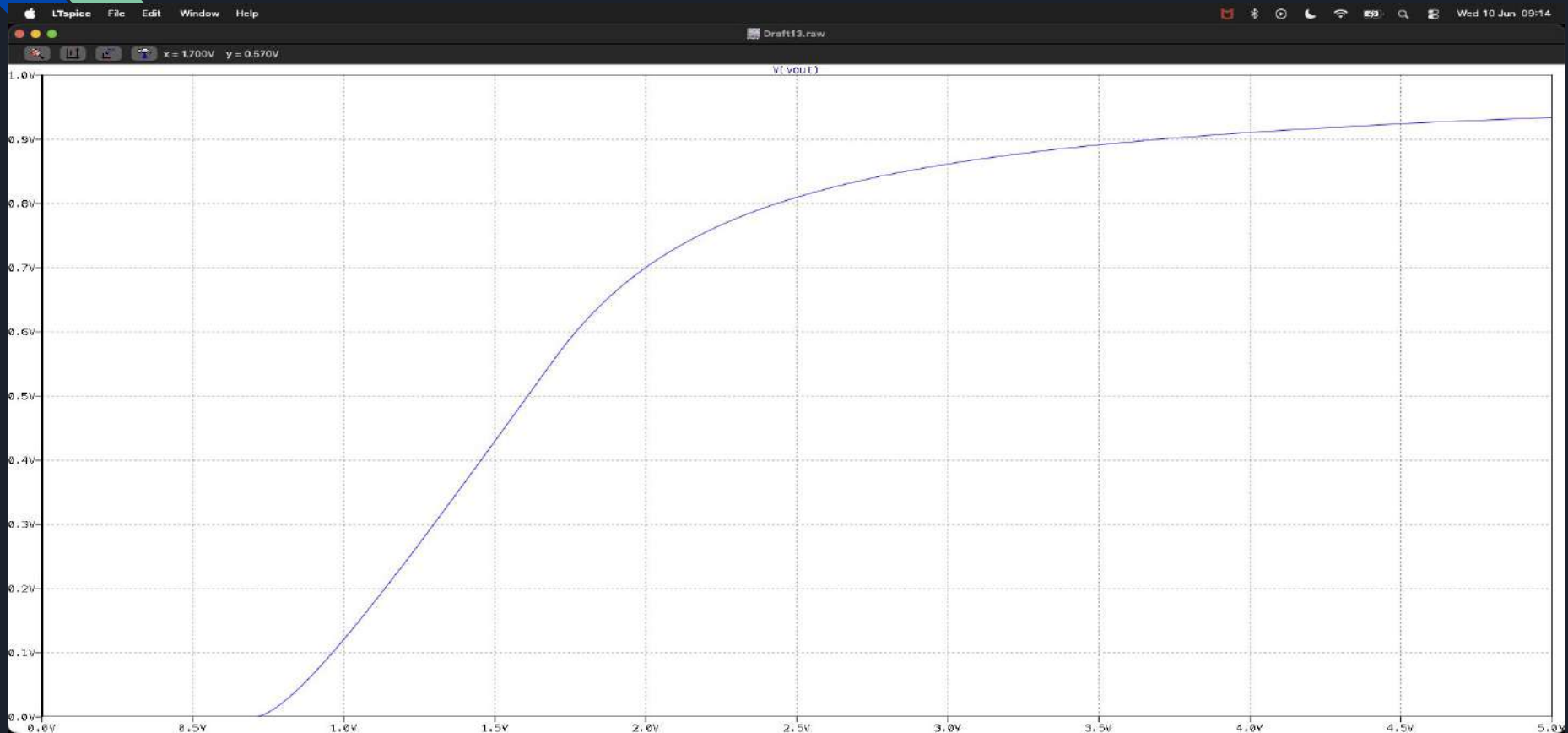
V_{out} will never reach 1V as there will be some voltage drop across MOSFET



LTSPICE Schematic



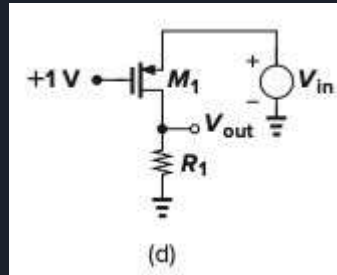
Simulation Results for Vout



Problem 2.7-d

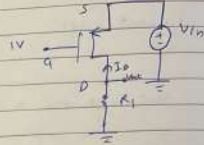
2.7. Sketch V_{out} as a function of V_{in} for each circuit in Fig. 2.49 as V_{in} varies from 0 to V_{DD} .

Diagram -



Handwritten Solution

27.4)



$$V_{gs} = 5 - V_{in}$$

$$V_{ds} = V_{out} - V_{in}$$

For PMOSFET to turn on

$$V_{gs} < V_{th}$$

$$5 - V_{in} < 1.8$$

$$V_{in} > 1.8$$

Once MOSFET turns ON, it enters saturation region

$$V_{ds} < V_{gs} - V_{th}$$

$$V_{out} - V_{in} < 5 - V_{in} - 1.8$$

$$V_{out} < 1.8 \rightarrow \text{saturation}$$

$$\therefore V_{out} > 1.8 \rightarrow \text{triode}$$

for $V_{in} > 1.8$ and $V_{out} < 1.8$

$$I_D = -\frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{gs} - V_{th})^2 = -\frac{1}{2} \mu_p C_{ox} \frac{W}{L} (5 - 1.8 - V_{in})^2$$

this is valid till $V_{out} = 1.8$

$$V_{out} = -I_D R_1 = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} R_1 (1.8 - V_{in})^2$$

$$V_{out} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} R_1 (V_{in} - 1.8)^2$$

$$\therefore V_{in} = 1.8 + \sqrt{\frac{2 \times 1.8}{\mu_p C_{ox} \frac{W}{L} R_1}}$$

after this point MOSFET enters triode region

for $V_{out} > 1.8$, $V_{in} > 1.8 + \sqrt{\frac{2 \times 1.8}{\mu_p C_{ox} \frac{W}{L} R_1}}$

$$I_D = -\mu_p C_{ox} \frac{W}{L} \left[(V_{gs} - V_{th}) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

$$= -\mu_p C_{ox} \frac{W}{L} \left[(1 - V_{in} + 0.8)(V_{out} - V_{in}) - \frac{(V_{out} - V_{in})^2}{2} \right]$$

$$= -\mu_p C_{ox} \frac{W}{L} (V_{out} - V_{in}) \left[\frac{1}{2}(1 - V_{in}) - \frac{(V_{out} - V_{in})}{2} \right]$$

$$= -\mu_p C_{ox} \frac{W}{L} (V_{out} - V_{in}) \frac{1}{2} [2(1 - V_{in}) - (V_{out} - V_{in})]$$

$$= -\mu_p C_{ox} \frac{W}{L} \frac{1}{2} (V_{out} - V_{in}) [3.6 - 2V_{in} - V_{out} + V_{in}]$$

$$I_D = -\mu_p C_{ox} \frac{W}{L} \frac{1}{2} (V_{out} - V_{in}) [3.6 - V_{out} - V_{in}]$$

$$V_{out} = -I_D R_1 = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{out} - V_{in}) [3.6 - V_{out} - V_{in}]$$

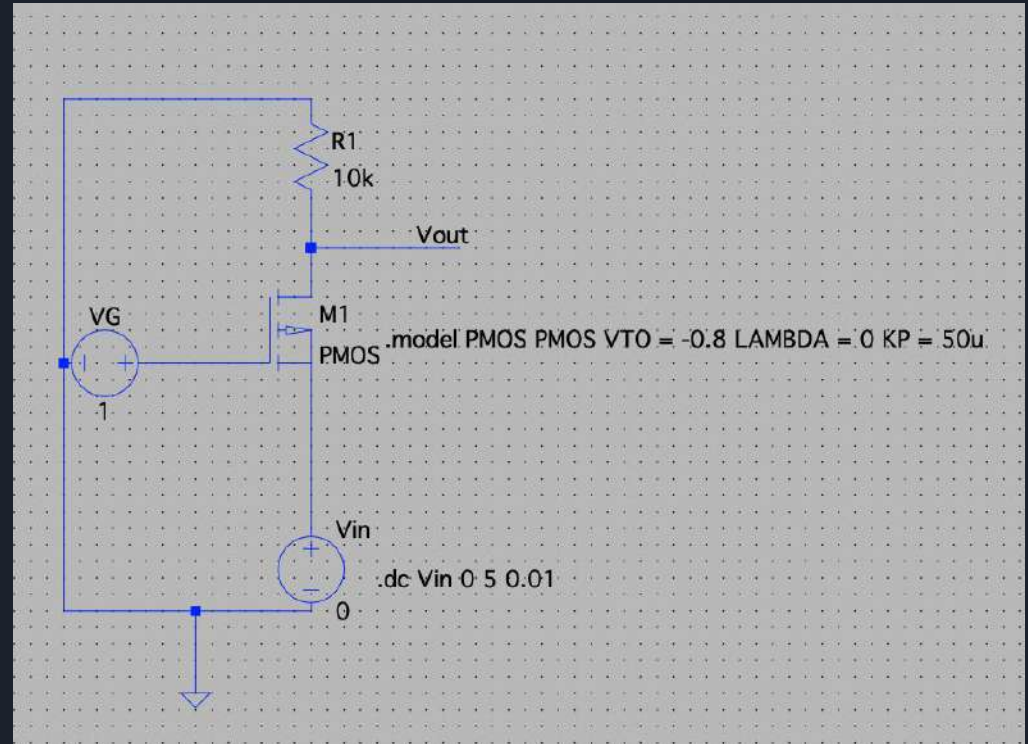


LTSPICE Schematic

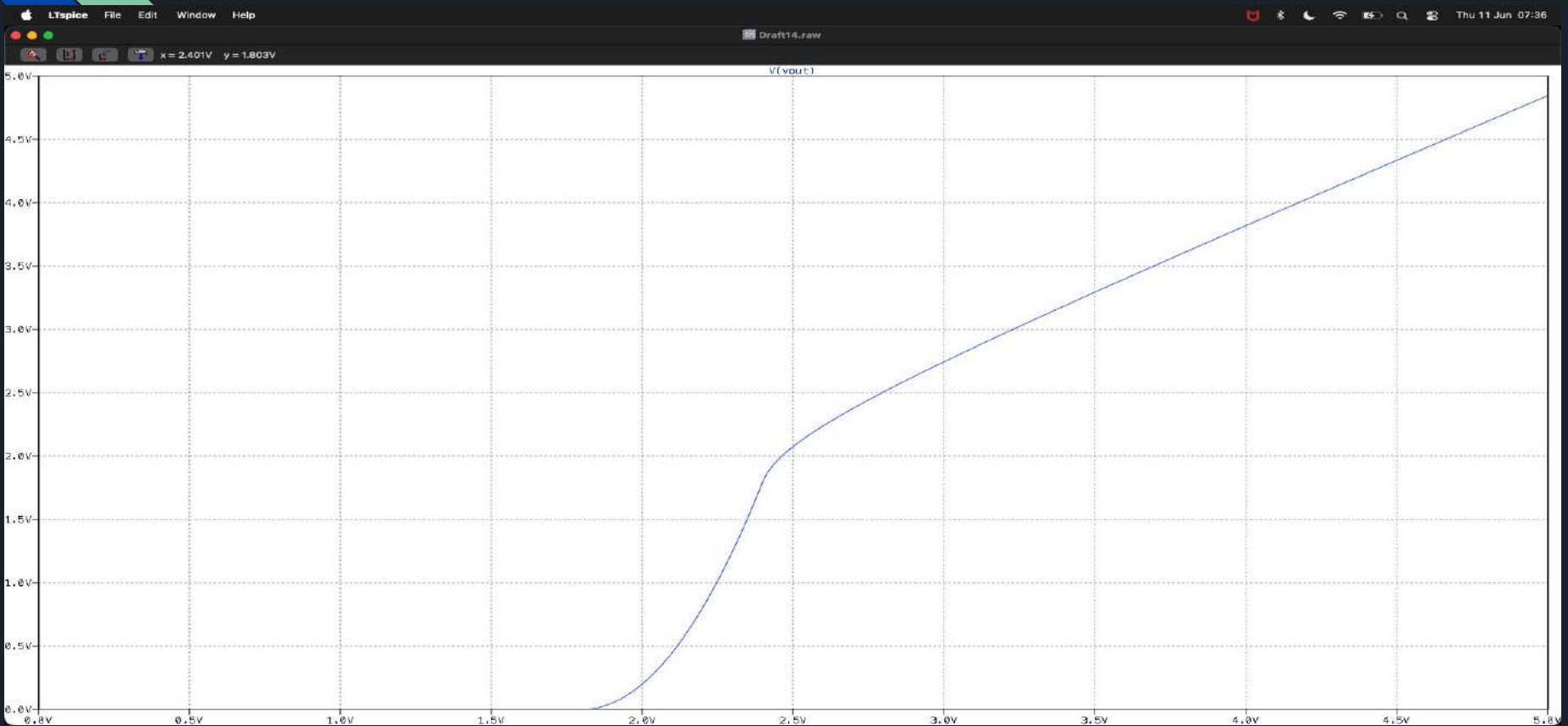
Assuming $\mu_{pox} = 50 \mu A/V^2$, $W/L = 20$, $R_1 = 10k$

transition point = $1 P + \sqrt{\frac{3.6}{\mu_{pox} W R_1}}$

$V_{in} = 2.4V$



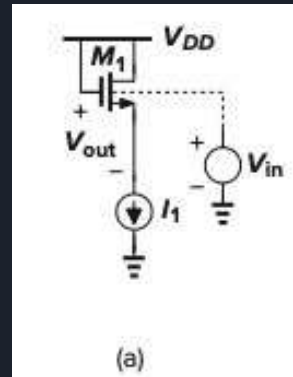
Simulation Results for Vout



Problem 2.8-a

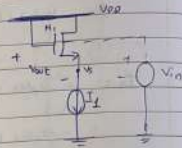
2.8. Sketch V_{out} as a function of V_{in} for each circuit in Fig. 2.50 as V_{in} varies from 0 to V_{DD} .

Diagram -



Handwritten Solution

28/4/24



$$V_{GS} = V_{DD} - V_i$$

$$V_{GS} = V_{DD} - V_i$$

$$V_i = V_{DD} - V_{out}$$

$$V_{GS} = V_{GS} = V_{out}$$

$$V_{GS} = V_{DD} - V_{out} - V_{in}$$

The MOSFET is always in saturation
as $V_{GS} > V_{th}$

$$\text{and } I_D = I_{D0} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 = \text{constant}$$

$$V_{GS} - V_{th} = \text{constant}$$

$$V_{in} = V_{GS} + V_{out} - V_{th} = 0.9 + V_{DD} - V_{out} - V_{in} - 0.9$$

$$\text{Assuming } 0.9 + V_{DD} - V_{out} - V_{in} > 0$$

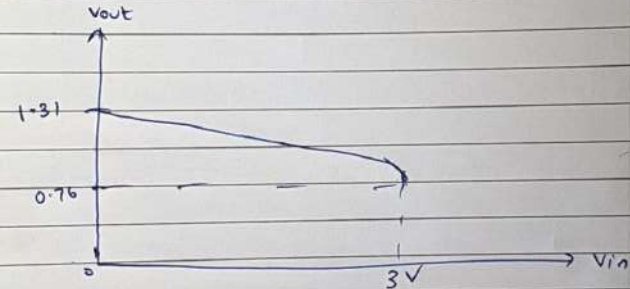
$V_{out} + V_{in} < 0.9$ [all points on the graph V_{out}, V_{in} must satisfy this]

∴ For current to be constant

$V_{GS} - V_{th}$ has to be constant

$V_{out} - V_{th}$ has to be constant

as $V_{in} \uparrow$, V_{GS} ↓ ∴ V_{GS} has to decrease, i.e. V_{out} has to decrease



Calculations

$V_{out} = 1.31V$, for $V_{in} = 0V$

$V_{out} = 0.76V$, for $V_{in} = 3V$

Assuming $2\phi = 0.9$, $\gamma = 0.45$, $I_1 = 1mA$, $V_{th0} = 0.7$
 $\mu_{n0} = 200mA/V^2$, $W/L = 50/0.5$

$$V_{ov} = \sqrt{\frac{2I_1}{\mu_{n0} \times W/L}} = 0.316V = \text{constant}$$

$$V_{ov} = V_{gs} - V_{th}$$

$$V_{gs} = V_{ov} + V_{th}$$

$$V_{out} = 0.316 + V_{th}, \quad V_{th} = 0.7 + 0.45 \left(\sqrt{0.9 + 3 - V_{out}} - \sqrt{0.9} \right)$$

$$V_{out} - 0.316 = 0.7 + 0.45 \left(\sqrt{3.9 - V_{out}} - \sqrt{0.9} \right)$$

$$V_{out} - 1.016 = 0.45 \left(\sqrt{3.9 - V_{out}} - \sqrt{0.9} \right)$$

$$V_{out} - 0.589 = 0.45 \sqrt{3.9 - V_{out}}$$

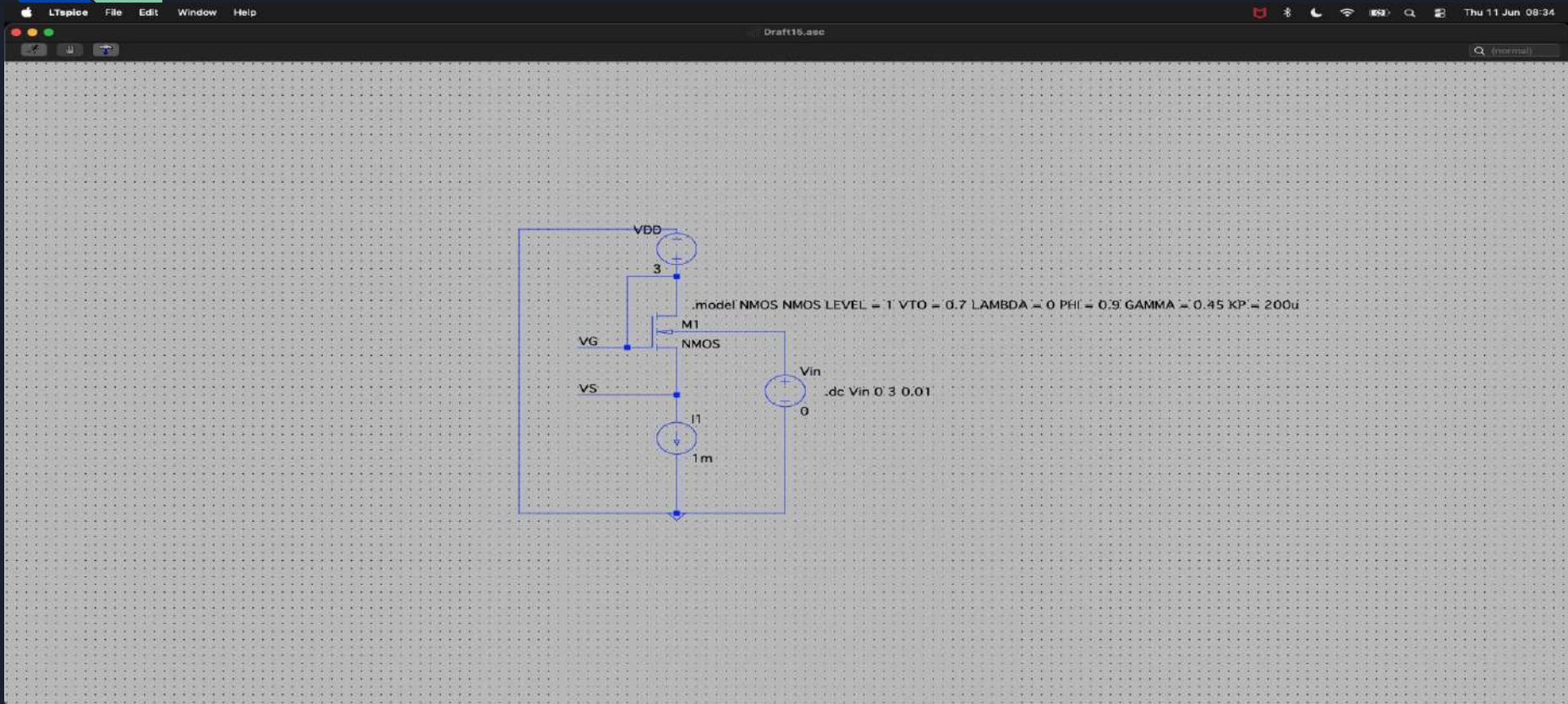
$$V_{out}^2 - 2V_{out}(0.589) + 0.346 = 0.2025 (3.9 - V_{out})$$

$$V_{out}^2 - 0.9755V_{out} - 0.44375 = 0$$

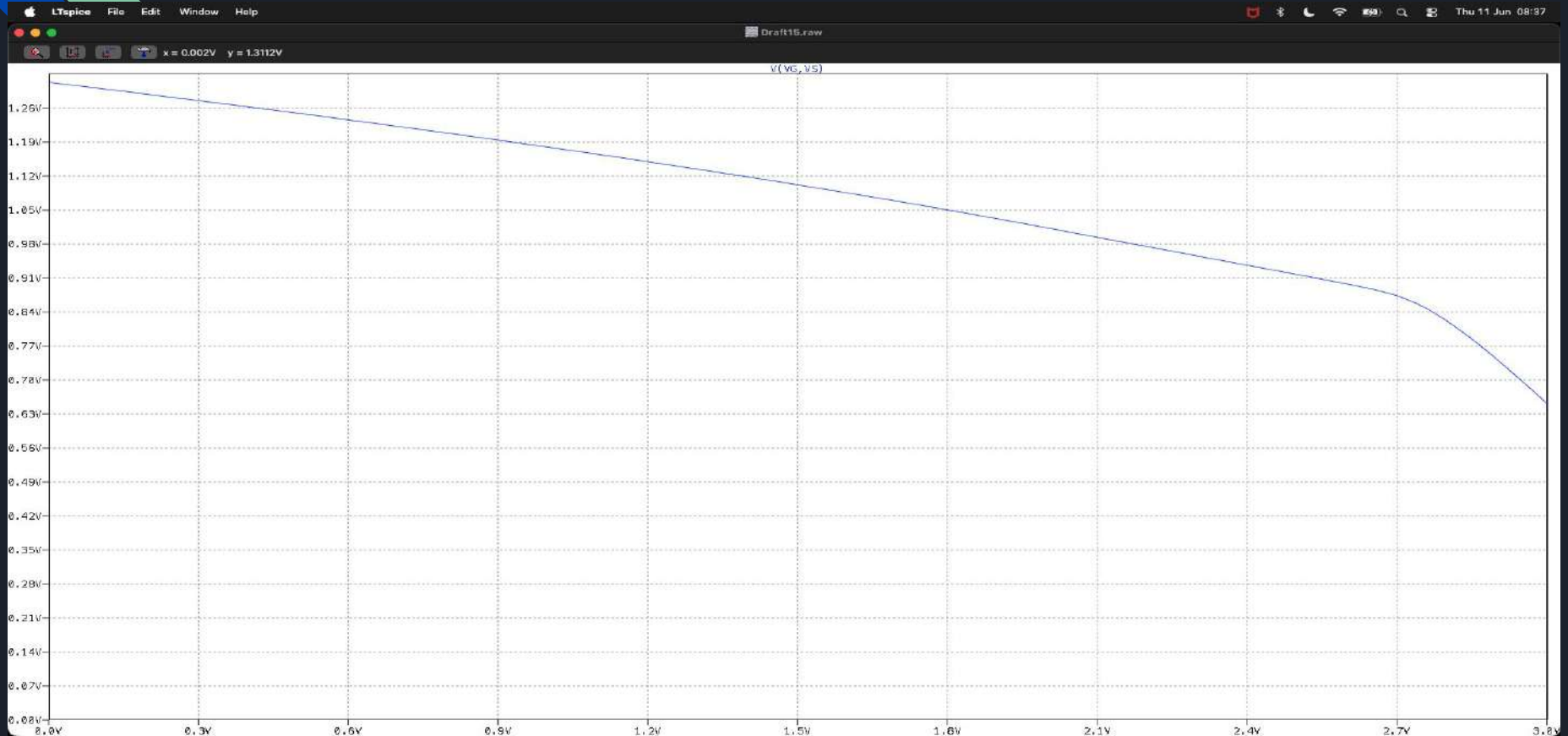
$$V_{out} = 1.313V \mid V_{in}=0V, \text{ similarly for } V_{in}=3V$$

$$V_{out} = 0.76V \mid V_{in}=3V$$

LTSPICE Schematic



Simulation Results for Vout





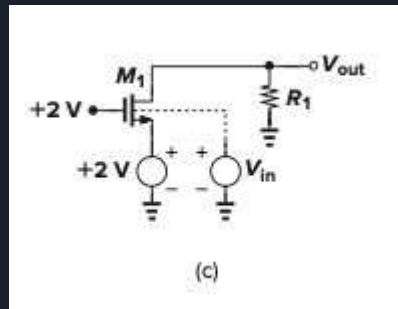
OBSERVATION

- As V_{in} approaches 3V, $V_{SB} = 3 - V_{out} - V_{in}$ becomes very small, causing the body-effect term $\sqrt{2\phi_F + V_{SB}}$ to change rapidly with V_{in} . Consequently, V_{TH} decreases more quickly, and since $V_{GS} = V_{TH} + V_{OV}$ with V_{OV} constant, $V_{out} = V_{GS}$ also falls more rapidly, making the slope steeper.
- The hand calculation uses the ideal square-law MOSFET and body-effect equations, which are valid only when the source-body junction is reverse biased. At $V_{in} = 3V$, V_{SB} becomes negative (forward body bias), so these equations break down, whereas LTspice uses a more realistic device model and predicts a lower V_{out} .

Problem 2.8-c

2.8. Sketch V_{out} as a function of V_{in} for each circuit in Fig. 2.50 as V_{in} varies from 0 to V_{DD} .

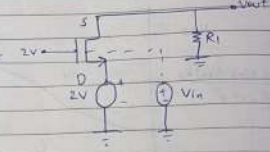
Diagram -



Handwritten Solution

2.4.6.4

Assuming $2\theta = 0.9$, $\gamma = 0.45$
 $\mu_{n0} = 100 \text{ cm}^2/\text{Vs}$, $W/L = 10/0.5$



Source and Drain exchange their roles

$V_{gs} = 2 - V_{out}$
 $V_{ds} = 2 - V_{out}$

$$V_{th} = V_{th0} + \gamma \left[\sqrt{2\phi_F + V_{gs}} - \sqrt{2\phi_F} \right]$$

$$2\phi_F + V_{gs} > 0$$

$$V_{gs} > -2\phi_F$$

$$V_{out} - V_{in} > -2\phi_F$$

Since $V_{gs} = V_{ds}$, MOSFET is in saturation region

$$I_D = \frac{1}{2} \mu_{n0} \frac{W}{L} (V_{gs} - V_{th})^2$$

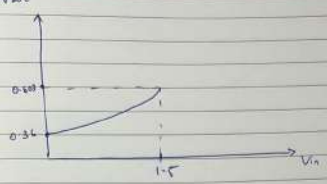
$$I_D = \frac{1}{2} \mu_{n0} \frac{W}{L} (2 - V_{out} - 0.7 - 0.45(\sqrt{2\phi_F + V_{gs}} - \sqrt{2\phi_F}))^2$$

$$V_{out} = I_{D1} = \int_{V_{th0}}^{V_{gs}} (2 - V_{out} - 0.7 - 0.45(\sqrt{2\phi_F + V_{gs}} - \sqrt{2\phi_F}))^2$$

When $V_{in} = 0$, we have $I_D = 0$ first
 $\therefore V_{out} = 0$

As $V_{in} \uparrow$, $V_{in} \downarrow$, $I_D \uparrow$, $\therefore V_{out} \uparrow$

V_{out}



for $V_{in} = 0$, $V_{out} =$

$$V_{out} = \frac{1}{2} \times 100 \times 10^{-4} \times 100 \times 10^{-9} \left(2 - V_{out} - 0.7 - 0.45(\sqrt{2\phi_F + V_{gs}} - \sqrt{2\phi_F}) \right)^2$$

$$V_{out} = 0.5 (1.3 - V_{out} - 0.45(\sqrt{2\phi_F + V_{gs}} - \sqrt{2\phi_F}))^2$$

$$V_{out} = 0.5 (1.727 - V_{out} - 0.45(\sqrt{2\phi_F + V_{gs}} - \sqrt{2\phi_F}))^2$$

$$V_{out} = 0.36$$

for $V_{in} = 1.5 \text{ V}$, $V_{out} = 0.525$

for $V_{in} = 1.5 \text{ V}$, $V_{out} = 0.503$

for $V_{in} = 2 \text{ V}$, $V_{out} = \text{invalid}$

$$V_{in} \leq V_{out} + 2\phi_F$$

$$V_{in} \leq V_{out} + 0.9$$

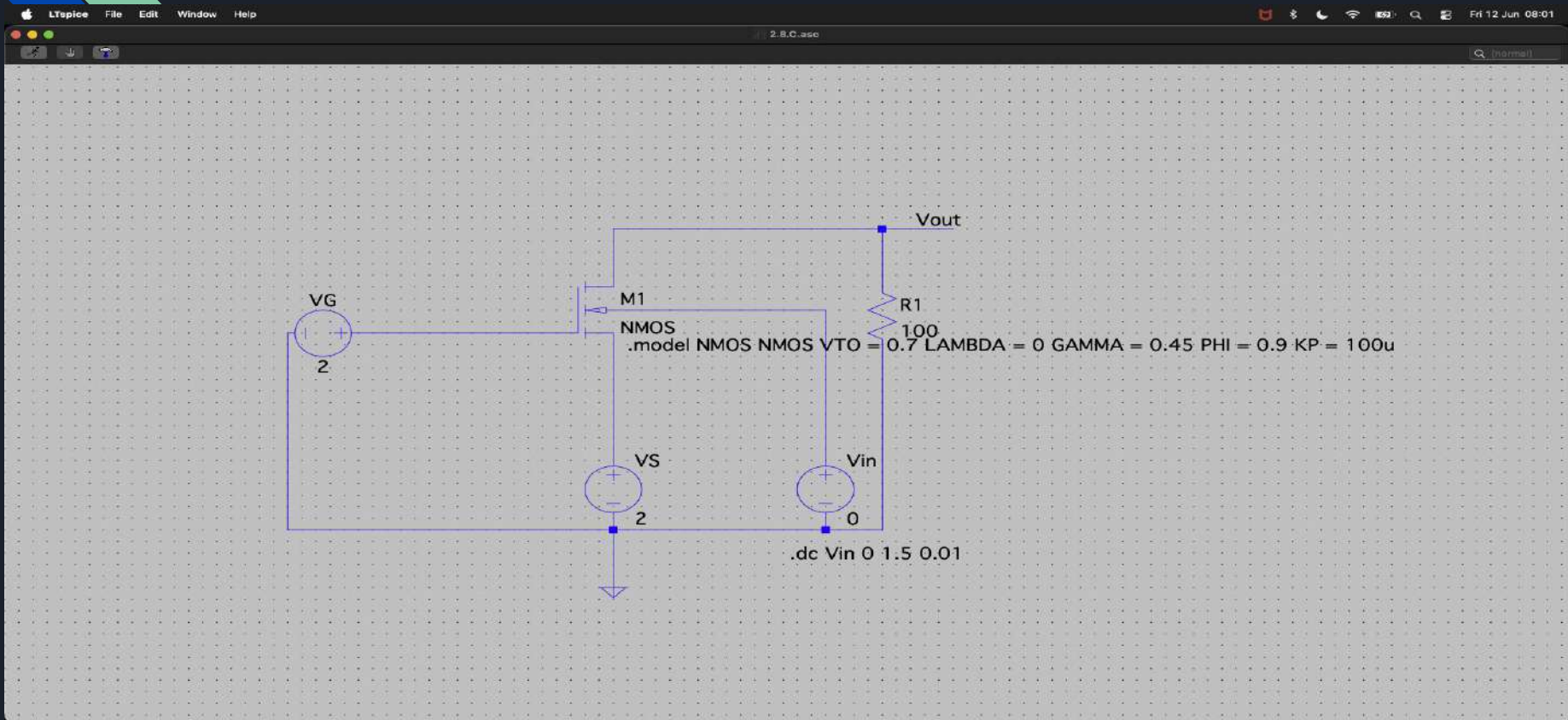
after $V_{in} = 1.5$, $V_{out} = 0.5$

$$V_{gs} = V_{out} - V_{in}$$

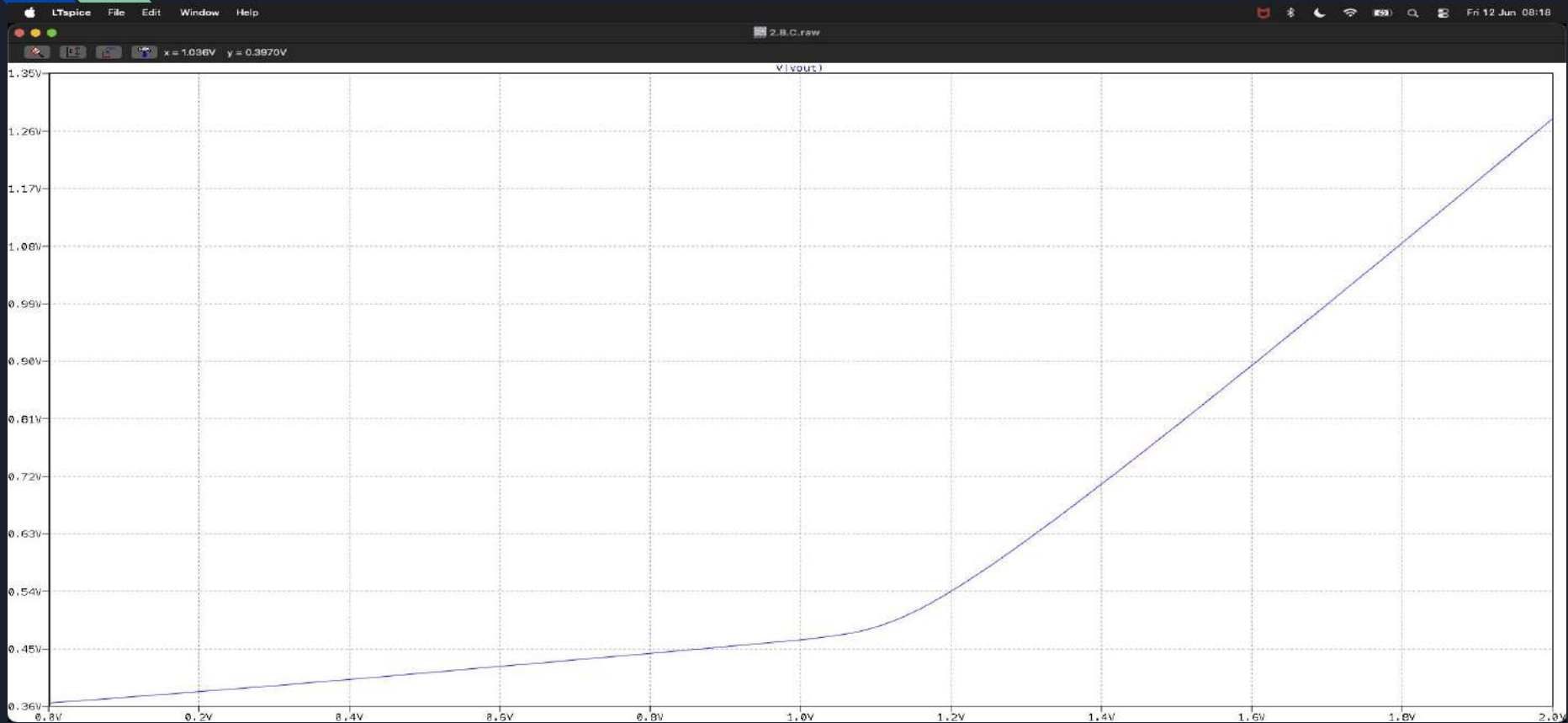
$$= -0.97 \text{ V}$$

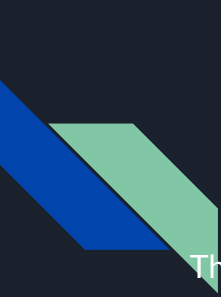
$$V_{gs} \text{ cannot be less than } -0.9 (-2\phi_F)$$

LTSPICE Schematic



Simulation Results for Vout





OBSERVATION

The mismatch between hand calculations and LTspice occurs because the analytical model considers only MOSFET channel current and body effect, while LTspice also includes the parasitic body-source diode.

- At $V_{in} = 0V$, $V_{SB} = V_{out} - V_{in} = 0.36V$, so the diode remains reverse biased and hand calculations match LTspice.
- At higher V_{in} ($\approx 1.5V$), $V_{BS} = V_{in} - V_{out}$ ($1.5 - 0.8 = 0.7$), which forward biases the body diode. An additional diode current flows: $I_{total} = I_D + I_{diode}$, causing deviation from the hand calculation.
- The sudden change in the LTspice curve occurs when the body diode turns on and clamps, $V_{out} \approx V_{in} - 0.7V$.

Conclusion: The difference is due to parasitic diode conduction neglected in the simplified MOSFET model.

THANK YOU

The background features a series of dark gray, three-dimensional rectangular planes that recede into the distance, creating a sense of depth. A bright green parallelogram is positioned behind the letter 'U' in 'YOU', and a bright blue parallelogram is located below it and to the right. The overall aesthetic is modern and minimalist.